1.0 **INTRODUCTION**

Agriculture means the production of crops, and the raising of animals to provide food, goods, and drugs to satisfy the needs of people.

The needs of people include

- Food
- Clothing
- Shelter
- Health
- Goods
- Money
- Employment/Work.

1.1 Food

This includes maize, sweet potatoes, onions, yams, tomatoes, pepper, cassava, garden eggs, cattle, plantains, sheep, goats, chickens, turkey and fish. These foods are eaten in various forms. Food crops may also be classified as cereals, legumes, roots and tubers, oil-bearing crops, fruits and vegetables.

1.2 Clothing

Clothing is obtained from cotton, skin, hides, and hairs. Cotton is obtained from the cotton plant. Animals supply us with hides and hair. Cotton is made into yarns, which are used to manufacture clothing. The Ghana Textile Factories produce various types of cloth.

Skin, hides and wool are from animals. Skin is obtained from small animals such as antelopes, goats, sheep, hares and rabbits while hides are supplied by large animals like cattle, elephants, buffalo and horses. Hides and skins are made into leather used to make belts, shoes and gloves. Hair is changed into wool which is then used to produce clothing including socks, pullovers and suits. They provide warmth and can be very useful in protecting the body from cold.

1.3 Shelter

Shelter provides protection against bad weather for humans, animals and crops; storage for farm products and resting places for humans and animals. Plant materials for shelter include wawa, odum, mahogany, bamboo, and thatch. Grass can be used for roofing.

1.4 Health

Farm products can be used to make drugs; neem as antimalarial and pesticides; pawpaw seeds for deworming; pawpaw leaves to soften meat; animal fats, for example pig lard, for drugs in factories.

1.5 Goods

Most goods in Ghana are produced from raw materials which are obtained from them.

1.6 Foreign Exchange

When we produce the goods we need in our factories from agricultural raw materials, we save money because we do not have to import them. We also produce cash crops like cocoa, pineapple, banana, pepper, yam, ginger, and export them to get money for our country. Some of the countries we export to are Britain, USA, Germany and China. The money we receive from our

exports enables us to buy things that we do not produce in Ghana.

1.7 Income

Income is the money we get within a certain time or period. When farmers produce their crops for the season, they sell some to get money. The money which they get for selling their crops during the growing season is their income. They need money to be able to pay for their hospital fees and pay school fees and buy clothing.

1.8 Employment

Employment is the regular work which one does for a living. The regular work of farmers is farming or agriculture. Other people whose work is in agriculture are tractor operators, herdsmen and farmhands. They work on the farm to earn a living. Agriculture offers work to about 60% of Ghanaians.

2.0 DEVELOPMENT OF AGRICULTURE

Agriculture has gone through many changes since early man started hunting and gathering.

2.1 The beginning of Agriculture

At first early man did not have farms from which he could get food. He moved from place to place for food. His food comprised wild fruits, honey and animals. His meat was obtained from animals he chased and killed during hunting. He also set traps and killed some wild animals which he used for food.

On his rounds, sometimes he caught young animals and brought them home to rear them. Some of the young animals later grew and were used as food. It is believed that the dog was the first animal early man caught and brought home. Later, the dog grew and became man's friend. It accompanied him when he went hunting for other wild animals. From these experiences, man learned to grow his own food and raise animals.

As time went on, early man collected seed and fruits from crops which had grown from what he had thrown away. When this pleased him, he increased the crops and cared for them. In this way, he was able to get food always without going to the forest to look for food.

The earliest farming tools were made from pieces of wood and stones. Later, bones of animals he had killed and eaten were used to make tools for the farm. These were used to clear woods, cut trees and skin the animals killed. The weapons were very simple. Clubs and stones were used to hunt animals.

The early farmers did not need to go about looking for fresh land to farm. They kept to one area of land. As the size of the family increased, the area of the land they needed to farm also increased. Gradually, man began to discover certain methods of growing his crops. When man first began to grow crops, he had no idea about growing them in rotation or manuring the soil as farmers do these days.

He suddenly found out that it was not satisfactory to continue growing the same crops on the same piece of land again and again. He thought that it was wise to allow his little piece of farmland to rest after cultivating it for a year or two. This area of farmland would become overgrown with wild grasses and other plants. This fallow period helped to add plant nutrients to the soil. This was the beginning of subsistence agriculture.

2.2 Subsistence Agriculture

Subsistence agriculture is the practice whereby farmers grow food to feed their families. Because the tools the early farmers used were simple, they could not clear large tracts of land to produce plenty of crops. They were only able to produce enough for themselves and their families. This type of Agriculture started with the early farmers but it still exists in many West African countries including Ghana.

In such farms, the tools the farmers use are cutlasses and hoes. Subsistence farmers have other jobs as their main jobs. Subsistence farmers raise animals but they do not house them properly. Their animals sleep in trees and on their compound at the mercy of the weather. The foods given to their animals are left-overs from the kitchen. Apart from this, these animals feed on grass and insects in the field. In many cases, they are underfed. Because of this their production is low.

Some subsistence farmers produce what they need for food plus a little more for sale. This started cash crop farming.

2.3 Cash Crop Farming

Cash crop farming is the type of agriculture in which crops are grown for sale. It is, however, not the only business of the farmer. A cash crop farmer may have other jobs. Cash crop farms are also called plantation farms.

This kind of farming began with the coming of the white man. On his arrival to certain parts of West Africa, the white man saw several tropical crops growing. He thought he could use some of them to feed the industries in his home country.

As a result, farmers started growing these crops and sold them to the white man. Cocoa and oil palm were some of the first crops grown for sale. Today, sheabutter, pineapples, citrus fruits, kolanuts, yam, banana, pepper and ginger are some of the crops grown for sale as cash crops.

Some of the countries which buy these crops from Ghana are Germany, Britain, USA and China.

2.4 Commercial Farming

Commercial farming is the type of farming where crops and animals are grown and reared on a large scale for sale. This type of farming is spreading all over West Africa including Ghana.

The farms are managed by farm managers, called entrepreneurs, who run the farms on business lines. They record their cost of production and sales to enable them to find whether they are managing the farm at a profit or loss.

Machines are used on these farms. In addition to the machines used, large quantities of fertilisers, herbicides and insecticides are used on such farms.

2.5 Development of Agriculture in Ghana

2.5.1 Hunting and gathering as part of agriculture in Ghana

Many farmers in Ghana still hunt for wild animals. Some farmers set traps even today to kill animals of all kinds for food. Men, women and children still go out in the night after rain with oil lamps to gather snails and mushrooms. During the dry season, hunters mainly from the northern parts are found all over the plains searching for animals in the bush, which they kill for food. Along the coast and in the forest areas farmers wait and collect coconuts, kolanuts and mangoes which have fallen from the trees.

2.5.2 Subsistence Agriculture in Ghana

Subsistence agriculture is still practised by farmers in Ghana. Farmers who practise subsistence agriculture sell a small amount of their crops but their main concern is to produce food for their families. They may have other farms which are mainly for cash crops which they sell to get money. Their farms are small in size and in most cases they do not use fertilisers. Because of this their yields are low. Some of them do not enter their farms after sowing until the crops are ready for harvesting. In such cases the farms are taken over by weeds and pests, which further reduce their yield.

Crops are not the only things produced in subsistence agriculture. Animals are also reared under subsistence agriculture. Animals produced under this type of farming are normally not housed. They may sleep in trees. They are given left-overs food from the kitchen and after that allowed to

run free-range. They eat whatever they come across, but mostly they eat insects. They rarely get water to drink and when they are sick they die without their owners knowing when and where they die. At times, they are taken by hawks or bitten by snakes in the field. Because of this their owners get very little out of them.

2.5.3 Cash crop farming in Ghana

Several crops are produced today for sale. At first the crops grown for sale were cocoa and oil palm. Today, crops like pine apple, kola, sheabutter, banana, ginger, citrus, pepper, and certain types of garden eggs are grown for sale. Britain, Germany, Russia, USA and Burkina Faso are some countries which buy cash crops from Ghana.

2.5.4 Commercial farming in Ghana

Commercial farming is receiving attention today in Ghana. Large areas are cleared to grow crops and also to raise animals for the production of meat and eggs.

Cassava, yam, ginger, pineapple, tomatoes, rice, pepper, onions, chickens, eggs, cattle, sheep and goats are some of the things they produce. Some of these products are sold abroad. Others are, however, sold in Ghana.

One special thing about commercial farming is that they are run on business lines. Records of income and expenditure are made. This helps the entrepreneur to know whether he is making a profit or loss. If he discovers that he is not making profits, he will immediately reorganise his business so as to make profit.

Because large areas are cultivated for growing crops in this type of farming, machines like tractors, combine harvesters and others are used to work on the farms. On such farms, large quantities of chemicals like fertilisers, herbicides and insecticides are used. Some of the commercial farms in Ghana and what they produce are given below.

3.0 **CROP ECOLOGY**

Crop ecology is the study of crops in their usual habitats or environments, and may be considered as consisting of two main areas:

a. Autecology is the study of the inter-relationships between a crop and its entire environment - that is, its aerial environment, and
b. Synecology which is the study of the structure or composition of the development and causes of distribution of plant communities.

Both branches of ecology are important from the point of view of an understanding of crop growth and distribution. In their growth, development and yield, crop plants require certain basic factors - these being nutrients, water, air, light, favourable temperatures and a suitable medium, these factors control the initiation and maintenance of vegetative growth, the transition from the vegetative phase to the reproductive phase, and the extent of reproductive growth.

Because the levels of performance of crops, that is their growth rates, their rates of development, and the quantity and quality of their yields or their products are the utmost results of the interactions between the plants and environmental factors, there is, therefore, the need also to have adequate knowledge of the genetical constitution and attributes of the crop species and varieties that are cultivated.

By virtue of their genetical constitution and other attributes, some crops are able to survive; that is, grow and reproduce, over a wide range of environmental conditions than others. The range of environmental conditions over which a crop can grow and reproduce successfully may be described as its Ecological Amplitude.

Thus, a crop with a wide ecological amplitude is one which is distributed or grown over a large portion of the globe; for example, maize which is grown from latitudes 55°N to 40°S; and rice which is grown from latitudes 45°N to 40°S. There are other crops which are grown over a smaller portion of the globe so that their distribution is more restricted. These crops are said to have a narrow ecological amplitude. An example is cocoa which is distributed between latitudes 20°N to 20°S, with the bulk of it produced between latitudes 10°N to 10°S. Another example is para rubber which is grown between latitudes 15°N to 10°S.

Differences in ecological amplitude arise because every crop species has a characteristic capacity for utilising the available resources of the physical environment. Crops are therefore influenced differently by environmental factors. Thus, the adaptation of a species or variety to a specific environment, or the ability of a species or variety to tolerate the environmental conditions prevailing in an area, has its foundation in genetics, so that the ability of a species or variety to respond favourably to the environment is governed by the laws of evolution, and, therefore, the range of tolerance, or ecological amplitude, is a direct result of the diversity of the species.

4.0 THE PHYSICAL ENVIRONMENT

The physical environment in which crop plants find themselves is a complex of factors which may be classified into four main groups.

a. Climatic Factors which include the general features of the regional climate and seasons, rainfall, solar radiation, temperatures, humidity and prevailing winds.
b. Physiographic Factors which include the general nature of the geological strata, topography - degree, direction and extent of slopes - exposure, elevation, ground water and the result of geodynamic processes such as erosion, silting-up and sand-blowing.
c. Edaphic Factors which include soil texture, structure, soil depth, mineral composition, available soil moisture, aeration and soil reaction.
d. Biotic Factors which result from the presence and activities as well as the influence of living organisms such as man, animals, soil-inhabiting macro- and micro-organisms in addition to other plants.

Nearly all the factors affect each other in rather complex relationships, in spite of this classification. For example, man can cause fire which can destroy the influence of other biotic factors, eg the presence of vegetation, and so accelerate certain geodynamic processes such as erosion caused by rainfall or wind. Similarly, climatic factors and physiographic factors together determine the nature of soils formed and so do influence important edaphic factors of the habitat. Again, elevation can substitute for latitude in terms of climatic conditions.

Some of the factors which constitute these four main groups affect plants directly. For instance, soil water, solar radiation, temperatures, soil mineral constituents and certain gases, influence the growth and development of crops whereas rainfall, wind, soil texture and structure and the activities of soil macro- and micro-organisms affect crops indirectly by influencing the intensity, duration and quality of the direct factors.

4.1 Climatic Factors

The study of climate concerns itself with the physical changes which occur in the earth's atmosphere, the lowest layer of which provides the environment for plant and animal life. In this context, agricultural meteorology should, therefore, be extended to include the study of the climate of the root zone of the soil. There are three branches of climatology which concern agriculture and forestry:-

a. Macroclimatology which is a study of the climate of a region as a whole, and is important in that it deals with the dominant climatic factors which determine the broad distribution of vegetation and the agricultural potential of a region, as well as the suitability of an area for specific crops or forms of land use eg oil palm and cocoa being typical tropical crops, oats and wheat being typical temperate crops, and steppes being for game and wildlife reserves, for example, the steppes of Kenya.

b. Mesoclimatology which is a study of the climatic conditions of a limited land area, eg a valley or top of a mountain range, where owing to aspect or topography, there may be considerable departures in one or more of the climatic factors from the normal of the region, for example, the southeast coastal region of Ghana.

c. Microclimatology which studies the modifying effects of vegetation or other physical

objects on the climate of a relatively small air space, vital in respect of insect habitats, conditions for the survival and spread of disease-causing organisms, spores, or the physiological reactions of particular plant parts eg. abortion of flowers on sympodial branches of cotton plants. Closely planted crops or trees may well have a modifying influence on their environment which may be either beneficial or detrimental. For example, shading of the soil reduces surface evaporation which is normally beneficial, but can reduce soil temperatures to a level which can suppress rate of nutrient and water absorption and this will be detrimental.

The most important factors to be considered under climate are rainfall, atmospheric humidity, temperature, light and wind.

4.2 Rainfall

It is a dominant factor in crop production and is, additionally, the only source of soil moisture in most tropical areas and in non-irrigated regions. Water is the main constituent of living plant tissues. It is the medium in which nutrients are transferred from the soil to the roots, translocated to the shoot and distributed to where they are needed. Its transpiration cools the plants when ambient temperatures are high. It is a universal solvent and plays an important role in metabolism, It is the source of hydrogen used photosynthesis to produce organic materials. In addition, it is needed by beneficial micro-organisms for their activities, for example, nitrogen fixation which provides plants with nitrates.

Although an important factor in crop growth and distribution, total rainfall, ie annual amount, is not as useful a criterion as its distribution during the year or season. For example, it is known that an annual rainfall of over 3000 mm, ie 100 ins. depresses cocoa yields and that the optimum annual rainfall is in the range of 1750 to 1280 mm. While knowledge of these annual totals is very helpful, it happens that their distribution during the year is even more useful as an index of expected yields, because it has been found that not only is cocoa distributed in areas where the precipitation from November to March (the usual dry season) is 300 mm or more, but also main-season yields are much higher if this November-March total exceeds 600 mm. At any one time the most useful component of rainfall is the effective rainfall which is given by the formula:-

EF = (Precipitation + Condensation) - (Evaporation + Run-off + Interception + Percolation beyond the root zone + Colloidal Absorption)

A proportion of the rainfall received at any time is effective in increasing soil moisture content. Generally, the cooler and more humid an area may be, the more effective a given amount of rainfall will be, because evapo-transpiration losses will be reduced. Similarly, a good soil structure which promotes more rapid infiltration, or with adequate soil conservation structures to reduce run-off, will have a great proportion of its rainfall retained as soil moisture.

Other effects of rainfall on crop production relate to erosion - an important cause of soil and land denudation, and the leaching of soil nutrients - mainly cations - to produce acidic soils - an important influence of rainfall on soil reaction, as well as its role in dissolving soil nutrients for plant uptake.

Important aspects of rainfall which are of relevance to agriculture are amount, duration and intensity, as well as its fluctuation or cyclical features from year to year.

4.3 Atmospheric Humidity

It is a measure of the water vapour in the air. The water vapour in the air when expressed as a percentage of that necessary to saturate the atmosphere at a particular temperature to cause dew or fog or rain is termed relative humidity. Thus the lower the relative humidity, the drier the atmosphere. The most important effect of relative humidity is that on evaporation and transpiration or evapo-transpiration - which is transpiration from plants plus evaporation from the surface of the surrounding soil - sometimes referred to as the Consumptive Use of Water.

Areas with low values of atmospheric humidity tend to lose a lot of water through evapo-transpiration. Thus, crops encountered in such areas would normally not survive unless they are deep-rooted so that they can tap the lower layers of soil for moisture, or they possess such morphological and physiological adaptations to such conditions in order to cope with such dry conditions.

Evaporation increases with temperature so that in warm or hot climes, it is the main determinant of the effectiveness of rainfall in increasing soil moisture content. Also, where evapo-transpiration rates exceed rainfall at a particular time, hardly any growth occurs. However, luxuriant growth leading to the establishment of forests, etc., occurs when evapo-transpiration rates are considerably less than rainfall received, so that, rainfall-evaporation ratios, usually used as an index of the external moisture relations of plants, are more than 100. It should be borne in mind that where the rainfall-evaporation ratio is 100% it means that at the time in question, rainfall and evaporation are equal. Thus. only xerophytic plants can thrive in areas where the rainfall is far below evaporation most of the time. Forests with big tree species and thick undergrowth occur where rainfall far exceeds transpiration losses.

4.4 Temperature

It is a dominant factor in the control of the growth and reproductive development of crops. All crops, in fact, have cardinal or critical temperatures below and above which their life processes virtually cease. Thus, strictly, every crop species has its own minimum and maximum temperatures. In between these critical temperatures are the optimum temperatures at which rates of growth and development are fastest and yields are highest. These minimum, optimum, and maximum temperatures depend on the crop variety, age, and condition of the plants and on other environmental factors, eg. relative humidity and light.

Generally, because there is very little, if any, biological activity in living tissues below 0°C and above 50°C, most crop plants have these as their critical temperatures. With tropical crops, minimum temperatures are usually around 10°C, and optimum temperatures usually lie between 30°C and 38°C during the day, and 20°C to 28°C during the night.

Above the minimum and up to the optimum, the rate of biological activity in plant tissues rises with increasing temperatures. (Within suitable temperature ranges, growth increases with a rise in temperature, often in approximately proportional relationships corresponding to the Van't Hoff-Anheimer law for mono-molecular chemical reactions ie a doubling in activity or growth for each 10°C rise in temperature). After the optimum the rate of biological activity declines with further increases in temperature up to the maximum. Thereafter, growth ceases owing to the destruction of hormones and enzyme systems, as well as some metabolites at the high temperatures.

Also, as temperature increases, the moisture requirements of crops increase and unless evapo-transpiration rates are balanced by increases in soil moisture content, hence increase in water uptake, then the plants first wilt and eventually die. Thus the two most important environmental factors which determine the length of the physiological growing season of crops are temperature and soil moisture content, and crops have to adjust (or adapt) themselves to fluctuations in these environmental factors in order to grow and reproduce in the climatic (or ecological) zones where they are found.

Another effect of temperature is that higher soil temperatures had to increase in the solubility and activity of mineral nutrients in soil water, as well as the rates at which these are absorbed by plant roots.

The responses of plants to changes in temperature are termed thermoperiodic responses, an important among these, with respect to both maximum (day) and minimum (right) temperatures, are:- the breaking of dormancy vernalisation (subjecting seeds or seedlings to a period of cold treatment in order to bring about their capacity to flower, or accelerate the onset of ripeness-to-flower), the rate of leaf output and expansion, leaf shedding, variation in the number of days to flowering, and respiration rates. Other factors, principally evapo-transpiration and light, usually interact with temperature in inducing these responses in plants.

It should be borne in mind that where most of these responses are due to seasonal variation in ambient temperatures, there are other effects which are occasioned by daily thermoperiodicity, and that since growth in many plants occurs mainly at night, night temperatures are important in the determination of growth as well as in development.

Another influence of temperature is that it determines the rate of the reactions which constitute the Dark Phase of photosynthesis and lead to the incorporation of CO_2 into organic material.

4.5 Light

The term, light is described as visible radiation which is that range of the electro-magnetic spectrum (0.39μ to 0.76μ) which affects or is perceived by the human eye. It would be pointed out, though, that other bands in the spectrum - near ultra-violet down to 0.32μ and very near infra-red up to 0.8μ - are also important in their effects on plants, and that the range of the visible band of the electro-magnetic spectrum which directly influences the process of photosynthesis is 0.4 to 0.7μ in wavelength.

Photosynthesis is the process by which all crop plants convert energy from the sun into chemical energy stored in organic materials which are used for growth. Thus, increase in the dry weight of plants, including their yield of edible dry matter, results from photosynthesis. Aspects of light which are important with regard to plant growth and development are its intensity, duration, and quality.

In general, light intensity is greatest at the equator, decreasing towards the poles. Light intensity can, however, be decreased by clouds and mists and the canopy of existing vegetation, thus leading to tall plants receiving the maximum benefit of the in-coming solar radiation. Typical plant processes influenced by light intensity are the photo-chemical processes which include photosynthesis, respiration (photo-respiration) and the formation, accumulation and activity of chlorophyll.

The duration of light, or the hours of sunlight (from sunrise to sunset), sometimes referred to as the effective length of day, also affects many crop plants. The duration of daylight varies from a nearly uniform 12-hour day at the equator to continuous light or darkness throughout the 24 hours for parts of the year at the poles. Thus, latitude and time of the year in relation to equinoxes are important determinants of the effective length of day. However, the extremes for any place depend on latitude. At about 50°N, which is roughly the northern limit for maize production, the extremes are about 16 hours and 8 hours. At 30°N, the range is about 14 hours to 10 hours. At 15°N, it is 13 hours to 11 hours. For Kumasi (6° 43'N), it is roughly 12.5 hours to 11.5 hours.

The responses of plants to the changing length of day, on the photoperiod, are referred to as photoperiodic effects and of all these effects including dormancy, seed germination, internode elongation, branching, tuberisation, leaf abscission and pigmentation, the most important in the control of flowering.

Flowering is a photoperiodic response to the timing of light and darkness. Every plant flowers only, or flowers most rapidly, when illuminated more than or less than a certain number of hours per day, this being known as its Critical Daylength. On the basis of this particular response which marks the transition from vegetative to reproductive growth, that is, whether a meristematic apex becomes floral (i.e. produces flower primordia) rather than vegetative (i.e. produces leaf initials), crop plants can be classified into three groups:-

a. Long-day plants which require on illumination longer than their critical daylength for flowering eg. barley and spinach.
b. Short-day plants which require illumination shorter than their critical daylength in order to flower eg. the tropical cereals - maize, sorghum, rice and millet and soyabean.
c. Day-neutral plants which are unaffected by, or indifferent to, the changing photoperiod eg. tomato and cotton.

In photoperiodism, it is interesting to note that with short-day plants, even a short interval of light during the night may prevent flowering, and that flowering in long-day plants is hastened by very short exposure to light during the night. Therefore, the length of the night (dark period) rather than the length of the daylight period is critical in these responses. Again, in these reaction of plants, the intensity of the light is not as critical as the period of exposure. Thus exposure to light at intensities which may be too low to be effective in photosynthesis can easily influence the control of flowering at such low intensities. The implication of these photoperiodic responses is that when placed in an unfavourable photoperiod, plants may not flower at all, and thus continue to grow vegetatively, or may flower at a time when their vegetative growth, or the development of their photosynthetic system, has not been adequate.

It should also be pointed out that the effective length of day can interact with temperature in its effect on crops, and that an example of a modification by temperature of a photoperiodic response is shown by sorghum (guinea corn). Quick maturing varieties of sorghum grow bigger in cooler areas if the day are long than in warmer areas where temperatures are more favourable.

The quality of light (spectral composition) is also important in photoperiodism. For example, red light of about 0.66μ in wavelength is most effective in inducing flowering in long-day plants, or preventing it in short-day plants, but far-red light of about 0.73μ in wavelength nullifies these effects when given soon after exposure to red light.

Phytochrome is a blue, proteinaceous pigment which is present in all angiosperms (flowering plants) - in seeds as well as seedlings - had been implicated as the major, if not the only, pigment involved in photoperiodism. Phytochrome is known to absorb red light most. The changing photoperiods are perceived by the leaves when light strikes the leaves. The controlling mechanism which involves plant hormones and their induction of floral initiation is then triggered off by phytochrome in a series of reactions in which it functions as an enzyme.

4.6 Air/wind

The air around plants supplies them with CO_2 for photo-synthesis. In the processes leading to photosynthesis, CO_2 from the surrounding air enters the leaf through the stomata, is dissolved in the water surrounding the leaf tissues, and diffuses into the chloroplasts which, when illuminated, incorporate it in a series of reactions into organic materials rich in energy. The surrounding air also supplies oxygen used in respiration to release energy from organic materials in plants for various purposes including growth, cell differentiation and absorption of nutrients and water from the soil.

Nitrogen in the atmosphere and in soil air can be made available to plats in usable forms by the action of lightning and by the activities of nitrogen-fixing microorganisms.

Contaminants or pollutants in air also affect crop plants. For example, CO resulting from incomplete combustion of fuels derived from carbonaceous materials is poisonous to all living organisms. SO_2 causes blotches between leaf veins. Fluorides cause chlorosis or death of leaf margins and tips. Thus, where these contaminants tend to prevail, only these plants that show some degree of tolerance to these gases are usually found there. Apart from these effects, the air around plants, moving as winds, can influence plant growth and distribution in the following ways:

a. by causing physical or mechanical damage to parts of plants eg. tearing away or breaking off of leaves and branches;
b. by causing lodging of whole plants;
c. by accelerating evapo-transpiration (which increases in proportion to the square root of mind velocity);
d. by disseminating pollen grains and seeds.

4.7 Other Factors of the Physical Environment

The remaining groups of factors, that is, physiographic, edaphic and biotic, in addition to socio-economic consideration - all of which have interacted to determine the present patterns of distribution of crop plants - are best considered together because, unlike climatic factors, their components do not usually influence crop growth, development and distribution directly.

5.0 GENERAL PRINCIPLES OF CROP DISTRIBUTION

After consideration of all these environmental factors and their effects on plants, it is now possible to draw attention meaningfully to the following principles which relate to the growth and distribution of crop plants and explain the rationale behind the ecological approach to agronomic investigations as well as underscoring the importance of phenological studies in crop research.

a. Crop distribution is primarily controlled by the distribution of climatic factors of which rainfall, temperature, light and wind have the most influence.
b. Crop distribution is secondarily controlled by edaphic factors. This control is secondary because soil formation itself is controlled by climate.
c. Every crop species or cultivar is able to exist and reproduce successfully only within a definite range of climatic and edaphic conditions, and this range represents the tolerance of the species or cultivar to external conditions.
d. Species or cultivars with a wide range of tolerance for all factors of the environment are likely to be widely distributed.
e. Species or cultivars may have a wide range of tolerance for one factor and a narrow range for another, hence when conditions are optimum for one factor, the limits of tolerance may be reduced with respect to another factor.
f. The period of reproduction is critical for the survival of the species, and environmental factors must not be limiting.

It should be quite clear, then, that in spite of genetic stability, differences in the distribution of climatic and edaphic factors can cause variation in the times taken by crop species or varieties to reach the different developmental stages during their growth. The rhythm of succession of these developmental phases is controlled by climatic conditions and, if data on the times of occurrence of these stages/phases are correlated with the distribution of the relevant climatic parameters, it is then possible to deduce not only the climatic requirements and limits of climatic tolerance of various crops, but conversely, the suitability of any area for the production of any crop.

The study of the occurrence of the successive developmental phases of a crop is referred to as phenology. Phenology is, therefore, an integral part of the process of determining environmental reactions of crops and ecological optima as well.

Ecological optimum refers to the highest degree of plant functions, culminating in yield, in relation to environmental factors. Any area with comparatively high yields of a crop and a low yield variation indicates stability of production and approaches the ecological optimum. High annual or seasonal variation in yield over a period of years indicates the presence of one or more limiting factors and a poor adaptation of a crop to the prevailing environment.

5.1 Origin and Spread of Crops

Since the domestication of crops by man on his emergence as a cultivator from nomadic practices, there have been considerable movements of crops from place to place both within and between continents by 4 major means.

a. Crops have spread through trade and, therefore, along trade routes eg. spices reached Africa, Southern Europe and the new world through the early Spanish and Portuguese traders who established trading posts in SE Asia, including India. By the same means, mangoes reached West Africa but through the slave trade.

b. Crops have spread as a result of migrations and conquests, eg. the early Malawi introduced bananas to Madagascar and E. Africa when they reached there in about 500 AD. Again, typical Asian crops such as oranges and sugar cane were introduced into S. Europe when the Arabs stormed there at the height of the Islamic conquests.

c. Crops have spread through explorations and work in experimental gardens, eg. Columbus took sugar cane and citrus to the Caribbean (West Indies) on his second voyage between the Old and New Worlds in 1493, and on his way back, carried maize and sweet potato to Spain. In less than a century i.e. by the middle of the 16th century, maize had reached China under similar circumstances. Also, Captain Blight took the breadfruit, Artocarpus altilis, from Tahiti to the West Indies in 1793 to be tried in a botanic garden at St. Vincent where it became established, and from where it spread to other places in the west. The Royal Botanic Garden at Kew, England and the Singapore botanic garden cooperated in the spread of rubber in SE Asia.

d. Crops have spread through other means, notably drifting of propagules on the sea. For example, the coconut is believed to have drifted across the Pacific from Polynesia to the West Coast of Panama.

5.2 Some important reasons for the success of such introductions have been:-

a. Adaptation or genetic adjustment of the various crops to the new environment.

b. The ease with which the introduced crop can be integrated into the existing agricultural system either as a food crop or as an industrial or export crop.

c. A rapidly increasing demand for the crop as reflected in the attractive prices offered.

d. Availability of planting material

e. Availability of land, labour and capital

f. Improved methods of cultivation, harvesting and farm processing as demonstrated and recommended by research station and extension staff.

g. The major pests and diseases which attack the crops at the centres of origin are usually not present in the new areas and it takes time, sometimes years, for the pests and diseases to reach the new areas, or for local pests and diseases to cause serious damage.

h. Political stability, leading to long-term policies on government subsidies and other forms of assistance in the production and marketing of the introduced crops.

i. Enthusiasm and intelligence of local farmers and planters who can supplement production of large-scale farms set up by government and entrepreneurs.

Following from all these, the distribution of tropical crops and crops in general has been such that the main areas of maximum production are often far removed from the centres of origin. Centres of origin are usually located by means of investigations which include cytological studies, as well as the location of areas with many related wild types or the greatest diversity of type of each crop plant. It is not uncommon that the environment in the new areas is more ideal for the crops that it is in the original homes.

5.3 Plant Introductions

The introduction of plants from one country to another is a very common practice in which hardly any regard is given to the scientific principles to be considered if these introductions are to be organised and carried out properly. The process should take into consideration what are referred to as "agro-climatic analogues". The determination of these climatically and agriculturally analogous areas is based on the principle that a given species or variety tends to be similar in its phenological behaviour, i.e. with regard to such weather-affected responses as times of flowering and fruiting, in areas of similar climatic and latitudinal conditions, these analogues being areas

that are sufficiently alike in those features affecting crop growth and development to offer a reasonable chance for success of establishment of plant material transported between them.

The climatic parameters to be compared in determining the analogues are: mean annual, seasonal and monthly rainfall, mean monthly day (maximum) and night (minimum) temperatures, precipitation/evaporation ratios, latitude, lengths of frost-less periods, solar radiation income, evapo-transpiration rates and the duration of daylight in the course of the year. In addition, soil and physiographic factors must be considered.

This matching of geographical areas on the basis of purely physical data must then be verified by biological means such as provided by the phenological records of similar plants growing in the areas being matched. Plants used for such biological verification should, if possible, be pure-live varieties or clones in order to ensure that genetical, attributes are held constant. If the phenological reactions of the cultivars are more or less similar in the climatically and/or latitudinally analogous areas, then this constitutes evidence that there is a biological check that the areas are agro-climatically analogous.

Table 1 Centres of origin and maximum production of various crops

Crop	Centre of Origin	Area of Max. Production
Maize	Central America	U. S. A.
Cocoa	South America	Ghana and Ivory Coast
Coffee	Ethiopia	Central and South America
Rubber	South America	Malaysia
Soyabean	North East Asia	U. S. A.
Groundnut	South America	India and China
Citrus	South East Asia	U. S. A.

6.0 **AGRO-FORESTRY**

Agro-forestry may be thought of as any agricultural system that combines trees with crops and/or animals either spatially, i.e. simultaneously, or sequentially. Thus, it is a system of land use and resource-management, as well as a productive system which, in contrast to monoculture, gives multiple outputs of food products and wood products.

It should be emphasized that not only should it be seen as a food and wood production system, but also that as a system of land use and resource-management, it can be applied in various ways, for example, water or erosion control on sloping land, stabilizing conditions in a catchment area, and reclaiming degraded or denuded lands for sustained productivity. It is worth noting, therefore, that the three integral key aims of the system are:-

a. increased and sustained food production
b. judicious forest management
c. improved and/or conservation of the resources of the land/soil

Essentially, then, the components of the system should have as objectives the following:-

a. diversification and optimum combinations of species per unit area of land,
b. permanent soil production,
c. continuous restoration of soil fertility,
d. the control of weeds, pests and diseases.

It is against this background that any recommendations or research programmes in agro-forestry must be evaluated for effectiveness.

7.0 SPACING OF CROPS DURING PRODUCTION

Plants in a community compete with one another for nutrients, water and light. To be able to compete successfully, they require space for their shoots and their roots. Generally, though years of selection and adaptation, agricultural plants have developed the ability to withstand competition from plants of the same species. In spite of this ability of crops to grow together in a community, it is essential that individual plants should be properly spaced, since their maximum yield can be expected, after rectifying nutrient and moisture deficiencies, only by allowing the individual plants enough space so that their canopies and roots can develop fully and function efficiently.

Dry matter production depends on photosynthesis, and the greater the area of leaf surface exposed to sunlight, the greater will be the amount of dry matter produced. Any factor which tends to reduce the amount of sunlight reaching the leaves of a plant, for example, shading of leaves by others in a closely-planted crop, will therefore tend to reduce the amount of photosynthate produced and will adversely affect the biological yield of the plant.

The rate of root growth depends, among such other things as temperature, moisture and oxygen supply in the soil, on the amount of photosynthate the aerial parts, particularly the leaves, produce and translocate to the root system, and on the space available for the growth of the roots.

It should also be evident that for a higher crop yield, it should be necessary to increase the number of plants per unit area to obtain the maximum possible use of the resources of the soil, including any added fertilizer, in addition to the fullest possible exploitation of the in-coming solar radiation. Obviously, when the essential growth requirements are not limiting, the yield from a hectare with 25,000 plants will be lower than that from a hectare with 35,000 plants. Thus, while providing enough space for individual plants, efficient use of space and the resources of the soil must, at the same time, be ensured by judicious spacing.

By field trials conducted over several seasons, it should be possible to determine the optimum ranges of planting distances for various crops under various environmental conditions. It should be pointed out that there are no fixed spacings for crops. Crop varieties, growth habits (eg. erect, semi-erect or prostrate varieties) local environmental conditions, farming systems, soils and cultural practices vary, and so must planting distances if high yields are to be maintained. However, in any locality, and under average conditions, certain planting distances can be generally applicable, and these are used to compute plant population densities. Examples are as follows:-

Groundnut
Spacing: 30.5 cm x 30.5 cm $\quad \frac{10000}{30.5 \times 30.5} = 10.8/m^2$

Tobacco
Spacing: 122.0 cm x 91.5 cm $\quad \frac{10000}{1.22 \times 0.915} = 9{,}000/ha$

Oil palm
Spacing: 9m x 9m $\quad \frac{10000}{9 \times 9} = \frac{1000}{81} = 125/ha$

A mere statement of the plant population density, though useful, is really inadequate because it does not give any indication as to how the plants are arranged on the ground. Thus, the spatial arrangements or plant rectangularity must also be stated to give a complete picture of the plant stand.

Various rectangularities can be used to one's convenience. A 1:1 rectangularity means the plants are spaced equally between and within rows. A 2:1 rectangularity means the distances between rows are twice the distances between plants in a row, and so on.

7.1 Relationship Between Yield and Plant Density

The yield of a crop is not a linear function of the density of plant stand because after a wide range of density, further increases in plant stand bring diminishing returns. The relationship can be fairly well determined by planting a crop at a range of densities and plotting the yields against the densities. The characteristic features of a typical yield/density curve are:

a. over an initial increase in density, yield tends to increase linearly
b. yield reaches a plateau of more or less constant value over a range of densities
c. in some cases, there may be a decrease in yield from further increases in plant stand.

Generally, the total yield of dry matter or biomass produced per unit area of land tends to increase to an asymptote or a plateau with increasing density. However, the relationship between density and some special component, such as grain yield, may show a decline in yield at high densities.

It is apparent from the graph that first, the total shoot weight of maize reaches an asymptote at high densities. At the high densities, the stem and leaf weight tends to increase further. Thus, vegetative weight does not quite reach a plateau. However, ear weight declines after a certain intermediate density. In general, the reaction of crops to increasing density is such that the reduced weight of the individual plants at high densities is compensated for by the increasing number of plants per unit area.

At any stage that the number of plants does not adequately compensate for the declining yield per plant, then total yield declines. As the stand density is increased, so the proportional allocation of photosynthate to the various organs of the plant may change and the evidence is that it is seed or grain output or production which suffers first in this changing allocation of dry matter.

8.0 APPLICATION OF FERTILIZERS

The purpose of a fertilizer programme is to maintain or increase soil fertility nutrient content in an attempt to supply the amounts of crop nutrients which will result in such an increase in crop yields as will bring in maximum possible net returns.

There are various ways of maintaining soil fertility. These include the use of animal waste, green manuring, cover cropping and ploughing in plant remains such as mulches. The use of inorganic fertilizers is becoming increasingly important in tropical agriculture and is the fastest way to replace lost nutrients.

The major factors which determine the selection, rate and application of fertilizers are:-

a. Nutrient removal by crops:
Depending on the species, plant size which is a varietal characteristics, period of active growth and plant population density, crop plants remove various amounts of nutrients annually and for an efficient use of fertilizers, it is necessary to know the approximate amounts of nutrients thus removed. It is also very necessary to know the uptake of nutrients by crops. For example, legumes require relatively little N since they are able to fix some nitrogen for their use with the help of nitrogen-fixing bacteria, but maize requires large quantities of N relative to P and K.

b. Crop root characteristics:
Roots are the principal organs used for nutrient and water absorption and the rooting habits of crops, whether they are fibrous or the tap system, and the depth to which they reach are important. There are specific and varietal differences in the root characteristics of crops, and there are also differences in their feeding power which arise from differences in their CEC and the amounts of CO_2 produced during respiration.

Other differences are due to soil effects eg. low natural fertility and compact B horizons.

c. Soil characteristics:
Important soil characteristics are the amounts of plant nutrients already present in the soil, the extent to which they are likely to be available to the crop during the growing season, the predominant day minerals in the soil which largely determine the fixation capacity of the soil, the soil depth and physical characteristics which influence water movement and aeration through the soil and the past management history of the land or soil, the organic matter contents of the soil and the soil pH.

d. Method of fertilizer placement:
This is an important factor which relates to efficient utilization by the crop, prevention of salt injury to the crop roots and convenience of the farmer. The methods available are broadcasting, plough-sole placement, band placement, in-the-row application, top or side dressing and ring application. The method selected should be geared to the mobility of the nutrients within the soil and general considerations such as equipment available, times of emergence and flowering, rainfall characteristics and plant height.

Of the three major nutrients N, P and K, N is the most mobile, and moves very freely in the soil profile. The two major forms of N are NO_3^- and NH_4^+ and NO_3^- moves more freely than NH_4^+ which tends to be adsorbed on soil colloids.

P moves very slowly in soils and tends to be rendered immobile by precipitation by hydrous oxides, particularly those of Fe, Mn and Al, in acid soils, and by Ca in alkaline soils as well as being adsorbed and retained by days saturated with Ca. Ca phosphates where solubility decreases in that order.

K moves slowly in soils also, but is relatively more mobile in soils, although it tends to become adsorbed on the colloidal complex.

8.1 Nature of the fertilizer

Ammonium fertilizers, particularly Ammonium Sulphate, acidify soils. K fertilizers are usually neutral in their effect on the soil pH, while Calcium fertilizers tend to make the soil more alkaline. Calcium and magnesium, being metallic cations, move in the soil in about the same way as Potassium. The functions of the nutrients are as follows:-

a. Nitrogen is a major element in plant nutrition and metabolism. It is a constituent of all proteins, and proteins form a large and indispensable part of the living protoplasm itself, as well as of all enzymes which have important functions in essential life processes. It is also a component of chlorophyll and of certain vitamins and hormones. It is required for vigorous vegetative growth owing to its function in chlorophyll formation and subsequently in photosynthesis.

b. Phosphorus is a constituent of Adenosine Triphosphate (ATP) and Adenosine Diphosphate (ADP) which play a major role in metabolic energy transfer and in photosynthesis. P is therefore required for certain reactions in the plant which contribute to the vigour of the plant and to active root growth. Hastens flowering in plants and influences active reproductive growth.

c. Potassium is associated with the metabolism of plants; it is, for instance, required for protein synthesis - being needed in the coupling of certain amino-acids to form peptides from which proteins are synthesized. It is also known to be actively involved in the utilization of water by plants, synthesis of carbohydrates and their translocation within the plant.

d. Sulphur is taken up by plants as sulphates, is essential as a constituent of some amino acids eg. cystine, and some peptides which are needed for active protein synthesis and vigorous growth. Sulphur is also needed for active nitrogen fixation.

e. Calcium is necessary for nitrogen fixation and in soils in which Ammonium is the main source of Nitrogen. It is a constituent of cell walls, some enzymes and is needed for activation of other enzymes. Together with Magnessium, it plays a major role in the structural arrangement of chromosomes.

f. Magnesium is a constituent of chlorophyll and important in photosynthesis. It also strengthens all walls in plants. Apart from crop removal, other factors such as leaching, volatilization, fixation, uptake by soil microbes and erosion can cause serious reductions in the levels of availability of nutrients in the soil. Thus, seemingly adequate fertilizer application may not really supply the amounts of nutrients required by plants and they may show symptoms of nutrient deficiency which should be detected for remedial action to be taken.

8.2 Deficiencies

a. Nitrogen and Sulphur deficiencies may result in smaller, pale green (chlorotic) plants. Nitrogen deficiency causes a yellowing of leaves particularly along the veins. Sulphur deficiency appears in young leaves first and can cause death of the tips and margins.

b. Phosphorus deficiency causes stunted growth, the leaves become very dark green with purple tinges in places.
c. Potassium deficiency shows by a yellowing of leaves which develops into orange mottling and necrotic areas at the leaf margins and the blades of older leaves.
d. Calcium deficiency is more prominently shown by younger leaves which show a bleached appearance and may be deformed owing to disrupted cell division.
e. Magnesium deficiency leads to a bronze yellowing of leaves, older leaves being affected first. The veins remain green (unlike in N deficiency) but the leaf tissues in between become chlorotic.

Plant nutrients have usually been classified as macro- and micro- depending on the relative quantities required for effective functioning. However, of late, some authors have classified them into

a. Primary - major ones required in large amounts, N, P, K
b. Secondary - required in not so large quantities, Ca, Mg, S
c. Micro - required in much smaller quantities, since they are mostly required for enzyme synthesis needed in cell metabolism - Bo, Fe, Cu, Cl, Mn, Zn and Mo.

9.0 **FALLOWS**

When a piece of land is left uncropped after a period of cropping, then it is said to be under fallow. Such resting land can be kept almost completely free of all vegetation, in which case it is said to be a bare fallow, or it can be made to carry vegetation which may be vegetation which has sprung up naturally (natural fallow) or vegetation which has been purposely planted.

Continuously cropped land loses nutrients, the soil structure deteriorates and the land subsequently becomes erodible. Leaving the land under vegetation, therefore, has the purposes of restoring soil fertility through nutrient recycling, building up the soil structure and protecting the soil against erosion.

9.1 Natural Fallows

The processes which occur may be summarized thus:-

a. Forest areas

Annual herbs colonise the abandoned land first, and on supplanted later by perennial woody plants that shoot up from stumps left during previous land clearing and seeds which remain dormant eg. Musanga cercropoides.

b. Savanna areas

Annual grasses and herbs colonise the fallow area first, and are then followed by certain perennial grasses eg. Andropogon and hyperata which persist.

Under the fallow, accumulation of nitrates, phosphates sulphates as well as potassium, calcium and magnesium occur as a result of a slow but steady build-up of soil organic matter due to litter fall and decay of roots of plants in the succession. Thus, the roots of colonisers and invaders feed from the zone of the top soil and part of the sub-soil. These nutrients are retained in the plants and later returned to the soil surface through litter fall. Continuous washing by ram and decomposition of the litter ensure the nutrients being added to the top soil and root zone. Meanwhile, the death of plant roots also contributes organic matter to the root zone. This re-cycling process continues for years, leading to the build-up soil organic matter and nutrients for as long as the fallow remains and the re-cycling process is not broken.

In addition to this build-up, the improvement of soil structure and the protection of the soil against run-off leads to improved infiltration of water and an increase in the soil moisture content.

10.0 WEEDS - THEIR EFFECTS, BIOLOGY AND CONTROL

In agriculture, a weed, by definition, is any plant, usually herbaceous, on farmland or cropland which is of no economic value, and is undesirable because of its interference with crop growth and with normal farm operations and activities.

10.1 Effects

a. Reduction in crop yield

Weeds compete with crops for space and light above ground, and for nutrients, water and oxygen below ground. Their ability to compete with crops and so reduce their growth rates and yields depends on their number (weed population), height, total leaf area exposed to the sun, and the extent, particularly depth, of their root system. Some weeds are even known to use cultivated crops for support. The direct result of this competition r interference is a reduction in the yields of crops which may be slight or severe to complete, depending on the level of their infestation of crop fields. Crop yield losses of 20% - 50% are very common, and in some cases, weeds can wipe out a whole crop.

b. Reduction in crop quality and value

Weeds can reduce the quality and value of crops when their seeds or residues contaminate harvested crops, particularly small grains. This contamination reduces the purity, hence quality and value of harvested crops. In addition, some weeds are poisonous and can render a whole crop unsuitable for human or animal consumption. Some pasture weeds can import undesirable or objectionable flavour to animal products when consumed with pasture or in hay and silage.

c. Increase in costs of production

Weeds increase the cost of production of crops in that where they are many, additional efforts must be made in controlling them by cultivation or chemical control and, in some cases, cleaning the harvested crop. Also, weeds in irrigation ditches and water diversion or distribution channels can impair the efficiency of use of these structures and have to be removed at additional cost.

d. Harbouring of pests

Some weeds can act as alternate hosts to insect pests and disease-causing organisms. Where the weed populations are dense, and depending on the weed species, they can harbour small animals like rodents which can destroy crops.

e. Reduction in land values

The value of agricultural land can be reduced, sometimes drastically, by the mere presence of certain weed species which are noxious or known to be very difficult to control or eradicate.

10.2 Classification of weeds

a. By habitat

Weeds can be classified by means of their usual habitat. Thus, there are weeds

- Associated with water eg. Sedges, for example, Cyperus in irrigated or flooded rice, and Pistia (water lettuce) in water distribution channels.
- Associated with grasses/pastures eg. Tridax
- Associated with crops eg. Abutilon with cereals
- Associated with certain practices eg. Crotolaria in constantly disturbed areas like

paths and roads.

In Europe and America, it has been possible to associate certain weeds with certain soil conditions eg, acidic soils, calcareous soils, soils with high nitrogen content, and soils with loose surface litter.

b. By life cycle

Weeds can be classified by the duration of their life cycles. Thus, there are

- Annual weeds which complete their life cycle in a season and die eg. Digitaria.
- Biennial weeds which complete their life cycle in two seasons, generally using the first for accumulation of dry matter, and the second for rigorous reproductive growth, eg. Crotolaria.
- Perennials which grow and reproduce by seed or vegetatively in more than two years or seasons.

c. By means of propagation

Weeds can be classified by their means of propagation so that there are weeds which propagate themselves by

- Seeds, as with most grasses
- Rhizomes and stolons, eg. perennial grasses like Cynodon and sedges like Cyperus
- Tubers and bulbils eg. Solanum
- Stems and leaves eg. Cactus and Bryophylum.

This method of classifying weeds gives and indication of the persistence of weeds. Those that propagate by means of seeds, for example, produce large quantities of seeds which can remain viable in the soil for many years or can remain dormant for some time before germinating. Persistence can also be related to very effective means of propagation, eg. by use of underground parts and creeping structures.

10.3 Dissemination (spreading) of weeds

a. By natural means, including

- Explosive bursting and censer mechanism eg. leguminous seeds.
- Wind dispersal of winged seeds and plumed fruits, especially those of the family Compositae.
- Dispersal by water in the form of run-off, streams and irrigation, as for most aquatic weeds.
- Dispersal by animals, such as seeds with hooks which stick to animal coats, and attractive seeds or fruits which are carried by animals from place to place, or eaten so that they are deposited in another place after passing through the gut.

b. By human agency, particularly

- Use of impure seeds for planting eg. the small grains like rice.
- Dispersal in hay, silage, manures and composts of seeds that are unaffected by those processes and to remain viable.
- By implements when seeds stick to farm equipment and are transported about
- By farm operators through their clothing and boots.

10.4 Weed control

Ideally, weeds must be completely eradicated, but this is an unattainable aim since weeds must always appear first before being discovered and removed. However, certain measures can be taken to keep their numbers down and so reduce their effects on crops:-

10.4.1 Prevention:

By legislation which allows certain specified weeds to be removed by law, by seed certification and plant quarantine measures, it should be possible to minimize the spread of weeds by preventing them from reaching various places.

10.4.2 Control:

The aims in weed control are

a. To reduce competition from weeds in the early stages of growth of crops where their effects can be damaging

b. To prevent established weeds from multiplying and spreading by destroying them before they seed or by destroying the vegetative structures used for spreading control can be by

- Mechanical methods eg. hand-pulling, hoeing and cutlassing, and by the use of mechanised tillage implements.

- Biological methods through
  - Rotations to control weeds associated with crops.
  - Competition in which leguminous species eg. Centrosema can effectively smother certain grass weeds and control them.
  - Pasturing which can control dicotyledonous weeds, and particularly, when goats can browse off dicotyledonous plants and pigs can remove tubers, bulbs and rhizomes.
  - Use of insect pests, eg. using the insect Cactoblastis cactorum to control Cactus.
  - Burning to eliminate weeds of all species from fields.
  - Flooding which controls non-aquatic weeds, and is particularly effective against deep-rhizomed perennials at the time of sprouting.

- By chemical methods in which certain chemicals, classified by their mode of action as herbicides, are used to kill weeds or prevent their emergence. There are two main groups of herbicides.
  - Non selective which kill all plants they come in contact with, and may have temporary or semi-permanent effects.
  - Selective herbicides (divided into two groups) which are so called because they leave the crops relatively unharmed because the crops can detoxify the herbicides while the weed cannot and therefore is killed.
  - Contact or morphological which exploit morphological differences between crop and weed and effective against dicotyledonous weeds in monocotyledonous crops. For example, most dicotyledonous plants have large leaves with lower surface tension, and their apical meristems are usually protected and usually covered by waxy materials and the leaves have a high surface tension.
  - Hormonal or physiological in which action is through absorption and their

effect on cell metabolism. These are usually plant regulators which, in smaller quantities can stimulate and increase growth rates, but in the concentration used as herbicides they over-stimulate, the plants which consequently die or are prevented from growing and developing.

In their application, the weedicides or herbicides can be used at certain times relative to the crop life

a. Pre-planting when they are applied before planting/ sowing

b. Post-planting when applied after planting/sowing

With post-planting, they can be applied before the seedlings emerge - pre-emergence or after the seedlings have emerged - post-emergence.

11.0 TYPES OF FARM ANIMALS AND THEIR IMPORTANCE

We already know that Agriculture is the production of crops, and the raising of animals and their preparation for man's use. Several animals are raised in Ghana. We wish to examine the different types of farm animals in this country and their importance to our lives. Animals kept by Ghanaian farmers include poultry, sheep, goats, rabbits, cattle, pigs and fish.

11.1 Poultry

Poultry are birds which are kept for their meat and eggs. The term poultry is used for all birds which have a pair of legs covered with scales. They have feathers covering their bodies and a pair of wings. They lay eggs and sit on them to hatch their young ones. The types of poultry kept by Ghanaian farmers are the same as found throughout the world. These are chicken, duck, turkey, goose, and guinea fowl. Chickens are the commonest of all the types of poultry. They are raised in almost all homes.

11.2 Sheep

Three types of sheep are seen in Ghana. They are the forest types, the Nungua Blackhead and the Long-legged sheep.

a. The Forest Types

The forest types have short legs and are small in size. The rams (the adult males) have horns but the ewes (the adult females) do not have horns. Their colour is usually black and white, all black, brown or a mixture of the colours. They can survive any weather. They are common in the rain forest areas of Ghana like Ashanti and the Coastal areas.

b. The Nungua Blackhead

The Nungua Blackhead usually has a white body with a blackhead. It was developed from the forest type sheep by the University of Ghana's Agricultural Research Station at Nungua in Accra.

c. The Long-legged Sheep

They Long-legged sheep are very common in Northern Ghana. They have long legs and large ears. The rams are horned and the ewes are hornless.

11.3 Goats

There are two types of goats in Ghana. These are the West African Dwarf and the Sokoto Red.

a. The West African Dwarf

The West African Dwarf goats are common in Southern Ghana. They can stand many farm animal diseases but they do not like water. They have short legs but are meatier. Their colour may be a mixture of ash, black, brown or white or they can be of one colour. Adult goats of both sexes may or may not have horns.

b. The Sokoto Red

The Sokoto Red are common in Northern Ghana. They have longer legs. Both the ram and ewe have horns. They have short ears which are horizontal.

11.4 Pigs

The types of pigs one can see in Ghana are the Ashanti Black, the Landrace, the Large White and the Comborough. The Ashanti Black are the local types while the Landrace, the Large White and the Comborough are the imported (exotic) types. The Ashanti Black pigs are common in the

forest and coastal areas. They are small and short. They also have short hairs. The exotic types have longer hairs. They also grow faster and have huge bodies. Of the three exotic pigs, the Comborough has recently been introduced into the country.

11.5 Cattle

Three types of cattle are reared in Ghana.

a. Beef Cattle

These are the types raised for their meat. They have large nuzzles and wide heads. The kinds of beef cattle kept in Ghana are N'dama and the West African Short Horn (WASH). They have tight skin with short hair. They can resist the cattle disease known as trypanosomiasis or sleeping sickness.

b. Dairy cattle

They are raised for their milk. They have large, soft udders. The hind-quarters are wide open to contain the udders. Their necks are long and thin. The kinds of dairy cattle in Ghana are the Sokoto Gudali and White Fulani.

c. Dual-purpose cattle

The third type of cattle are the dual-purpose ones. They produce enough milk and good meat. Examples are the West African Short Horn and the White Fulani.

11.6 Rabbits

Two types of rabbits are kept in Ghana; the local and the exotic types. The local types are smaller in size than the exotic types. The exotic types come from Europe, America and France. Imported rabbits in Ghana include the California White, the New Zealand White, the Chinchilla and the Flemish Giant. On most farms where rabbits are raised, the local and the exotic ones are mixed so as to obtain some types which can produce good meat and also withstand the local climate.

11.7 Fish

Fish farming is becoming popular in the country because of recent campaigns by government. Many fish ponds are being constructed in almost every town and village. The types of fish being reared are tilapia and catfish (mud fish).

11.8 Other farm animals

Some animals are also kept on a small-scale by farmers. They include horses, donkeys, and dogs. Currently, some farmers are rearing ostriches.

12.0 IMPORTANCE OF FARM ANIMALS AND THEIR USES

Farm animals are important because they give us food, clothing, power, and manure. They are also used for sports, security and medicine.

12.1 Food

Food products from animals reared in Ghana are meat, eggs and fish. We obtain meat from poultry, sheep, goats, pigs, cattle, rabbits and, in some areas, horses and dogs. We obtain milk from cattle. From milk we produce butter, cheese, yoghurt and creams. In some other countries, milk from sheep and goats is also used. Poultry birds give eggs but chicken's eggs are commonly eaten. We get lard from pigs. It is good for cooking.

12.2 Clothing

Skin and hides from cattle, sheep and goats are used to make clothing and many types of leatherworks. People in Northern Ghana use leather to make various things for weaving and decorations. Some of the leather goods are talismans, whisks, bracelets, amulets, mats, foot pillows, sandals, shoes, purses, bags and drums. In some temperate countries, sheep hair (wool) is used to make woollen materials. Sheep and goat skins are used to make carpets. Rabbit fur is used for making clothing in temperate countries and for making decorations. The hairs of pigs (bristles) are used to make brushes.

12.3 Power

Bullocks, bulls and horses are used as sources of power. In the Northern part of Ghana, some people use the donkey as a source of power too. They may all be used to help with transportation and work on farms. Government institutions are being encouraged to use animals for ploughing. This has already been started at the University of Ghana's Agricultural Station at Nungua in Accra. The use of animals will save much foreign exchange on fuel and spare parts for machinery.

Animals are strong and can work harder, and for longer than tractors. We must therefore encourage their use especially in the North and the coastal areas where the land is level.

12.4 Manure

Droppings from the floor of the pen house, called litter, are often collected and added to the soil. They are called organic manure or farmyard manure. They are very helpful to vegetable plants. They contain most of the nutrients required by plants. When the animals are killed, their internal organs (offal) are added to prepare chemical fertilisers.

12.5 Sports

Some animals may be kept for pleasure, sports and to showcase culture. The horse in Ghana is kept for several uses. It may be kept for pleasure. Some people like riding on horseback for enjoyment. In the North it is traditional for chiefs to ride on horseback in state. The horse is also used for sports and in games, such as horse racing and polo.

12.6 Security

Dogs are used to keep watch over property in homes, on farms and to hunt for meat. Cats also kill pests like mice and cockroaches.

12.7 Medicine

Parts of animals are used to manufacture medicine and medical sutures for hospitals. Oil is obtained from fish to make cod liver oil.

I INTRODUCTION

Animal Husbandry is the breeding, feeding, and management of animals for the production of food, fibre, work, and pleasure.

Modern methods concentrate on one type of animal in large, efficient farming units that generate animal products at the highest rate of return for investment. Intensive husbandry conditions include large numbers of animals in small pens or cages, enriched feed, growth stimulation by various means, and vaccination against disease. Most of the world's domestic animals, however, are raised in small units under less efficient conditions and at lower rates of return.

Animals furnish about 28 per cent of the world's total value of agricultural products. They supply a much higher proportion of human food in the developed countries than elsewhere.

Traditional husbandry practices are closely associated with the degree of control needed over the animals that are kept and with the uses to which they are put. Most domesticated animals have multiple uses; for example, animals kept primarily for work also supply milk, meat, and clothing materials. The animals and their uses, however, are closely associated with the culture and experience of the people who care for them. In some regions of the world, cattle are not used in the production of food. Studies have shown that the work power, fertilizer, milk, and the fuel from dung that the cattle provide in these regions are more efficient animal products than meat. Analysis of other cultural practices has often revealed that animals are put to unexpectedly efficient use according to local circumstances.

Environmental influences such as climate also play an important role in the domestication and use of animals. Water buffaloes are used as draught animals in southern Asia, where they have adapted to the high temperatures and humidity, whereas horses, which thrive in moderate climates, were the principal draught animals in the temperate regions until they were replaced by tractors. Cattle from India that are acclimatized to hot and humid conditions are prevalent in the southern United States because they are better adapted to the climate of the region than are European cattle.

II DRAUGHT ANIMALS

Domesticated animals used primarily for work, transport, and leisure are widely distributed. They include the horse, mule, donkey (or ass), ox, buffalo, camel, llama, alpaca, yak, reindeer, and dog.

Modern horses are thought to have descended from one or more of three subspecies, including the tarpan, Przhevalski's horse, and the European forest horse. They are still used for draught, that is for pulling heavy loads, in many countries of the world. They are also used for controlling other types of animals, for carrying packs, and for riding for leisure and sport. The world population of horses is estimated at more than 60 million; approximately half are in North and South America and half in Asia and Europe. The 12 million or so mules in the world are fairly evenly divided among Africa, North and Central America, and South America. Of the 40 million or more donkeys, about half are in Asia and a quarter in Africa.

Camels, llamas, and alpacas are used as beasts of burden. Of the 17 million or so camels in the world, approximately three-quarters are found in Africa and the rest in Asia. The llama and alpaca are mostly limited to small areas in South America. Practically all water buffaloes are

found in Asia, and are used primarily as draught animals, although they have potential for the production of milk and meat. Oxen are also important draught animals in Asia and in some parts of south-eastern Europe. Dogs are used as pack animals in Alaska and Siberia and are also used to control sheep and other animal herds.

III SHEEP AND GOATS

Sheep are used for wool, meat (mutton and lamb), and to a small extent for milk. They are commonly divided into three types based on whether their wool is fine, medium, or coarse. Perhaps the first animals to be used in husbandry, they were domesticated in south-western Asia about 11,000 years ago, and approximately a billion are now widely distributed throughout the world, with the largest populations in Africa, South America, Asia, Europe, and Oceania. Commercial sheep farming is usually conducted on large tracts of land, divided into operational units containing 1,000 or more animals per unit. Sheep are also raised as a secondary enterprise on many small farms. They are well adapted to semi-arid regions and to land that is too steep or rough for the cultivation of crops.

Goats were first domesticated in the same region as sheep, and for the same uses, but 1,500 or so years later. They are about two-fifths as numerous as sheep and show a similar distribution pattern.

IV PIGS

Recent evidence suggests that pigs were domesticated about 9,000 years ago in several world regions simultaneously. Worldwide, there are now estimated to be more than 700 million. Approximately half are raised in Asia, primarily in China, Europe, Russia, and South America also have large numbers. Unlike most domestic animals, pigs are omnivorous and compete directly with humans for many foods.

V CATTLE

Modern cattle are usually divided into three types: beef, dairy, and dual purpose, an intermediate type used for both milk and meat. It is believed that cattle were domesticated about 8,500 years ago in south-eastern Europe, with south-eastern Asia a probable second centre of domestication. World cattle population is more than 1 billion, with half concentrated in South America, Europe, Russia, the United States, and India. Most beef cattle are raised on large tracts of land, but, following weaning, the young animals to be used for meat may be fattened in pens. Dairy cattle are managed in relatively large herds under intensive conditions near centres of dense population. Cheese, dried milk, and other specialized products, however, usually come from small farms with cows set out to pasture.

The term *poultry* mainly covers chickens, ducks, turkeys, geese, guinea fowl, and squabs. Each of these domesticated groups is descended from a closely related wild bird and was probably first developed in the areas where the wild bird was indigenous. World numbers are estimated at more than 6 billion chickens, with somewhat more than 100 million ducks and slightly fewer than 100 million turkeys. Estimates for other types are not available. Chickens are numerous in most regions of the world, while most ducks are produced in Europe and Asia, and most turkeys in the United States. In the developed countries, production units for meat and eggs are large and intensive, with individual birds housed singly in cages or housed together in large numbers.

VII OTHER ANIMALS

Other domesticated animals of importance include, in the cattle family, the yak in Tibet, the mithan in India, and the banteng in south-eastern Asia. The reindeer is important in the northern regions of Europe and northern Asia.

Although a few specialized farms have raised various animals (primarily mink and foxes) for their pelts, most furs are obtained from wild animals by hunting and trapping. In North America, muskrats and raccoons provide slightly more than half the total revenue. The former Union of Soviet Socialist Republics (USSR) was also a leading fur nation.

VIII CURRENT TRENDS

The major concern of animal husbandry today is the extent to which production can be maintained as the human population increases. The goal of research has been to increase the efficiency of the production of animal products by genetic selection and genetic engineering. For example, scientists have increased the size of sheep by inserting growth-hormone genes into embryos, and the same method can be applied to cattle and other animals.

Research is also being directed towards the development of animal strains in areas deficient in protein. Thus in South America the capybara is being utilized by Venezuela for its meat; it is taken from the wild in large numbers to lands devoted to cattle raising. In Peru, the government is encouraging the raising of guinea pigs for their food value, and elsewhere in Latin America research is being conducted into the farming of the iguana lizard.

Intensive methods for producing animal products include confinement of poultry in small cages, pigs in small pens, and sheep and cattle in small lots. Confinement leads to savings in labour, feed, and other production costs. Confinement in individual pens may also afford savings from improved disease control and better protection from predators. Such practices have been condemned as being cruel to the animals, and protective legislation has been advocated. Livestock and poultry farmers, on the other hand, argue that the animals are probably not under much stress because disturbed animals usually show sharp decreases in productive capacity.

Also at issue is the use of special additives, including hormones, antibiotics, vitamins, and other substances to increase growth or productivity. General guidelines state that any supplement must be proved safe and effective in the amounts used. The two most controversial growth additives are the hormone diethylstilbestrol (DES) and antibiotics. DES is now banned because in high doses it was found to be capable of causing cancer. Scientists who advocate the banning of antibiotics argue that resistant strains of bacteria in animals might transfer their resistance to bacteria that infect humans. The physiological limits of all domestic animals may eventually prevent further increases in yield. If the human population continues to expand and food supplies become scarcer, the proportion of food that can be used to feed animals will then decrease, and animal products will be in shorter supply and more expensive.

CURRENT STATUS OF AGRICULTURAL MECHANIZATION IN GHANA

Ghana has passed through the initial stages of importing tractors and mechanized equipment for agriculture and a fairly large number of farmers, especially those in the transitional zone and the savanna zones in both northern and coastal south, are very much aware of the potentials of mechanization, especially tractors. As a result three distinct levels of agricultural mechanization do currently exist in Ghana.

a. Small-scale farmers

These are farmers use traditional hoe and cutlass as the main form of farming equipment. Occasionally farmers in the forest areas with trees to fell may use such motorised equipment as chainsaws, but the cutlass and hoe remain the main farming implements.

b. Draught animal farmers

Group of farmers using draught animals and animal-drawn equipment to farm large units.

c. Farmers who use wheeled tractors and associated implements

This group cuts across all classes of farmers, but predominantly includes medium and large-scale farmers who use such equipment as combines alongside tractors to farm such crops as maize or rice in large fields under a mono-crop system.

Generally depending upon the agro-ecological zone and the type of crops grown, one may find only one or all three levels of mechanization co-existing in an area. The traditional cutlass and hoe farmer is found throughout the country in all the agro-ecological zones, but they are more predominant in the forest zones in Eastern, Western, Brong Ahafo and parts of Ashanti, especially among farmers producing such staples as plantain, yams, cocoyams, cassava, cocoa and oil palm.

Production in the middle transitional zone and also in the southern coastal savanna belt is based on a mixture of hoe and cutlass technology and tractor technology to produce such crops as yams, cassava, maize, rice, cowpea, groundnuts, vegetables, sugar cane and pineapples.

Mechanisation is most evident in the Guinea and Sudan Savanna zones in the Northern and Upper Regions of the country where all the three forms of mechanization technology of hand tools (cutlass and hoe), draught animal (mainly bullocks)and motorised (tractorization) co-exist and are operated side by side.

Across the country however, the general level of mechanization is very low, and Ghanaian agriculture is predominantly one of the cutlass and hoe. Mechanization is centred in rice, maize, pineapple and vegetable production, especially in the irrigated areas and in the Northern and Upper Regions, where it permits a more intensive use of the available land.

Post-harvest handling

Agricultural produce needs to be transported after harvest to processing and consuming centres. Postharvest handling of agricultural crops comprises two main operations: gathering and carting. While gathering is mainly manual, carting may be substantially mechanised. Manual gathering and transport are the main handling methods within the confines of traditional Ghanaian farms, where produce is gathered onto arms or baskets and pans and may be carried on heads to a central collecting point or a central threshing bay, either to await on-farm processing or transport to the village or the marketing centre. There is no mechanization of this operation at this level on

most typical Ghanaian farms. Crop losses especially among cereal such as rice paddy are known to be substantial.

Beyond the boundaries of the farm, the handling of produce comes with the general rural and farm transport system which is discussed later in this report. Suffice it to say that, in addition to head-loading and manual handling, push carts, carts hitched to draught animals, tractors and bicycles are used for carting in the rural area. For large quantities of produce carted over long distances, trailers hitched to tractors, pick-ups, trucks, vans and heavy duty lorries are employed.

Food processing and storage

Food processing

Even though food processing entails both primary and secondary processing, for the purpose of this work they will be considered together as one. Furthermore, processing can be described as either traditional or improved but all operations differ from crop to crop.

In order to enhance this narrative, the operations involved are treated under five main food crop groupings; roots and tubers, cereals, legumes, fruits and vegetables, as well as oil-bearing crops.

A. Roots and tubers

In Ghana, cassava, yams, and cocoyam are the main crops under this group. Of these, cassava is the most processed crop. It is processed into chips, *kokonte, gari, agbelima*, and starch.

Chips and *Kokonte*
Manual and mechanised chipping machines for the production of spaghetti chips are currently being promoted among processors. Larger chips and *kokonte* are still produced manually with knives and cutlasses. Transverse slicing chippers are being developed in the country, but this is still at the prototype stage.

Gari Production
Traditional *gari* production still depends on manual operations. Improved techniques of processing involve the introduction of mechanised grating and pressing of the dough. Wooden and metal graters of capacities upwards of 500 kg per hour exist in the *gari* production trade.

A manually operated cassava grater was under evaluation in the Volta region. Manually operated and engine- and motor-driven screw presses, as well as hydraulic presses, are in considerable use in cooperatives. Occasionally these services are provided by other entrepreneurs for a fee. Not withstanding, most small-scale processors still use the traditional stones to press grated cassava for *gari* making and the technology of the screw press is not widely used.

The frying of the *gari* dough is not mechanized and it is not uncommon to see women sitting behind unhealthy fireplaces doing the frying with big frying pans and spoons used to turn the product being fried. Some processors use improved smokeless stoves which reduce heat stress on the person doing the frying.

Agbelima
This is fresh cassava dough that is eventually mixed with maize dough in the preparation of *banku* and *akple*. Peeled roots are passed through fine toothed graters or burr mills (traditionally referred to as corn mills). In the remote areas, manual grating over perforated metal sheet or tin is still used. But in most villages, towns and cities most processors use motorized graters or burr mills.

Starch and Flour
Milling of fresh roots for the production of starch and dried roots into flour both benefit from the use of burr mills, although hammer mills are in use for the production of flour. In some villages however women still use the mortar and pestle technology to pound the chips into flour.

B. Cereals

Maize, rice, sorghum and millet are the main cereal crops, consumed in large quantities by both humans and animals in Ghana. All these crops are threshed after harvest using a combination of manual threshing, using threshing boxes, sacks and sticks, while small-scale mechanical threshers are used for medium scale production. Rice is hulled at the local mill with special hullers which are also used for de-hulling maize for further processing into flour, or are manually dehulled using mortars and pestles.

Maize is shelled with the aid of several small-scale shellers including flat wooden (NRI) and metal cone (AESD and SG-2000). Other rotary cone shellers which are either free-standing or table-mounted are in use; there is currently a project under way to collect and catalogue them for future dissemination. A table sheller (*Kichanja*) from Tanzania is also being evaluated for possible adoption in Ghana.

For all these cereals, cleaning is by winnowing which is done manually by tossing the mixture of crop and chaff into air streams by use of shaking sieves or baskets. Pneumatic mechanical powered systems are used in a few places for removing foreign materials.

The use of mechanical threshers, shellers and winnowers is considered to be still very limited and there is the urgent need to popularise existing technologies.

There are reports from the Western Region of itinerant maize sheller operators using the engines from mist blowers as the power source. Shelling is considered by most farmers as an area requiring urgent mechanical intervention in the maize production system.

Milling of maize, sorghum and millet is carried out with burr mill. In some rural areas, stone grinding and pounding in wooden mortars are still widely practised.

For feed formulation or compounding, mixers and mill-mixers are also available to the large-scale concerns. Capacities range from 500-1000 kg per batch.

C. Legumes

The most important operations are decortication or shelling. These operations are still predominantly done manually. A multi-purpose sheller-cum-cleaner has been developed for cleaning and shelling simultaneously. In the absence of such mechanical devices, the two operations may be carried out separately and manually. The level of mechanization is still low.

D. Fruit and vegetables

Domestic grinding using the traditional *apotoyowa* and pestle persists in most households. Others use various grinding stones in the homes. However, small specially designed motorised burr-mills are available for milling small quantities of pepper and tomatoes for the preparation of soups and stews. These are portable and are therefore moved from place to place, but they are more common among urban dwellers.

Small juicers have also been imported into the country either as single units or attachment to food blenders and confectionery production gadgets, but the general adoption of these mechanical aids, particularly among farming communities, is limited.

E. Oil-bearing crops

Coconuts, oil palm, shea nuts, groundnuts and soya beans are processed into vegetable oil. Production of coconut oil benefits from the availability of mechanical graters and screw presses during the conversion of fresh coconut meat and copra respectively.

Stripping, digestion and pressing operations in palm oil extraction are mechanised. Cracking of palm nuts for the production of palm kernel oil is also fairly mechanised with the development of palm nut crackers. Plate mills are used for breaking the roasted kernels down before the oil is clarified.

A paddle dough kneader is also used in northern Ghana in the production of shea nut butter and groundnut oil. A horizontal screw press is currently being used for the production of soya bean oil.

It is worth mentioning that all these mechanised operations have not altogether replaced all the traditional manual operations used over the years. They are, however, gradually replacing some of these traditional operations and introducing innovations for more efficient production. They have also not been adopted en masse for several reasons; the most important being appropriateness and availability at affordable prices.

Food preservation and storage

Only limited preservation of agricultural products actually take place in the fresh state. Instead, drying to enhance storability is practised very widely. Drying is done mostly by sun-drying with a sprinkling of solar drying. In some large-scale installations, batch and continuous mechanical dryers are in use for drying crops like maize, sorghum, rice, pepper and other vegetables.

The introduction of the narrow ventilated crib (FAO) in the south and drying patios in the north has revolutionised the drying of cereals, pepper, and cassava chips.

Storage of crops differs from place to place and from crop to crop. Due to the scarcity of wood, storage structures constructed from alternative natural and renewable resources are prevalent. Farmers therefore use grass, cereal stalks and mud to construct their structures.

In the south, where wood is readily available and other industrial materials exist, the storage structures are more durable and serve the purpose of keeping out the moisture. Among the various storage structures are aluminium, steel and ferro-cement units of various shapes and sizes for storing mainly cereals.

Some current developments for the cereals include structures of rhomboid shape made of plywood and aluminium which are raised above the ground to prevent rodent attack and ensure ease of unloading from the bottom.

Storage of yams

Apart from the cereals, for which MOFA/FAO has introduced the narrow ventilated cribs, yams, an important tuber crop in the diet of Ghanaians, are stored by farmers in the producing areas for purposes of food security and also to increase the market value.

There are various traditional storage structures which are identified by farmers. The five main structures are the *Zana* mat structure, burying in a pit, the *sigu*, the farm hut and under-shade storage.

There are minor variations in these storage types as one moves from a locality to another. For example the Zanamat structure and the Farm Huts are popular in the Kintampo and Tamale areas, while the Sign, which is structurally quite similar to the Zanamat structure, is common in the Bole and Jentilpe area. Burying and storage under shade techniques are most common in the Mandari area [PHU/MOFA-GTZ Post-Harvest Project: Survey On Postharvest Systems of Yams and Cassava, 1994]. The MOFA/PHU is working on these storage structures, and have a Demonstration Plot at Tamale.

Farm and rural transport

The movement of materials to and from farms is a very intricate and labour consuming task, and transport is considered a major constraint at the very heart of the Ghanaian agricultural system. Transport is needed for carrying both people and produce, and its absence due to lack of roads, as well as suitable vehicles, has meant that most people and animals have to transport themselves on foot.

In Ghanaian agriculture, the major difficulty is in the handling and transport of farm produce from the farm site to the market place or to the homes. The road from the village to the farm area is usually only a footpath, perhaps only 0.5 metre wide, with, as in most cases, only foot bridges across the streams and rivers. Consequently the main mode of farm transport system is by carrying oneself and produce on foot, and through head portage. Hand carts are in very limited use, and are generally confined near urban centres.

In the Northern and Upper Regions, bicycles and motor cycles as well as draught animals are being increasingly used for farm transport where they are available. Power tillers which have been recently introduced in some areas are also increasingly being used mainly for farm transport.

A special project, known as Ghana's Rural JMT Pilot Project, launched in 1988 with participation of the UNDP/ITDG/TCC and GRATIS, introduced the use of cycle trailers capable of carrying up to 120 kg in the Northern regions of the country. So far 600 of the units have been assembled and distributed to women user groups in villages in the trial areas. It has been the hope of the project to confirm the useful role that non-motorised transport can play in local farming communities, to enable general adoption in other parts of the country as an alternative cost-effective transport. The rate of adoption is yet to be assessed.

Crop production

Another important feature of farm mechanisation in the country is that generally there is a high degree of selective mechanisation, only specific farm tasks being mechanised, except in the highly mechanised large-scale commercial farms such as Ejura Farms.

Levels of mechanisation of specific farm tasks in crop production may be summarised as follows:

Land Clearing

Land clearing is mostly done with traditional methods using machetes, cutlasses, axes, hoes, rakes and forked sticks. The number of operations vary from slashing, tree felling, size reduction burning, cleaning and hoeing, depending upon the vegetation. In the forest areas there are situations when motorised chain saws are used in the tree felling and size reduction. On the commercial large-scale farms highly mechanised methods of land clearing, using bulldozers, angle dozers, rakes and rippers are used.

In the savanna zones, some farmers use herbicides for clearing. The equipment predominantly used by such farmers is the knapsack sprayer.

Seedbed Preparation

Land preparation is generally influenced by ecological and economical factors, including soil type, land use during the previous cropping year, type of crop to be planted, and available technology options.

The majority of Ghanaian farmers prepare their fields manually, using hoes of different shapes and sizes. Operations may involve hoeing, making mounds, ridges or beds. The short-handled broad-bladed hoe constitutes the main hand tool used. The traditional farmer hardly uses such implements such as spades, pick-axes and fork. Land preparation for yam cultivation consists mainly of making mounds, which is done manually. In general the level of mechanisation here is near zero.

In the savanna zones a fair proportion of farmers use bullocks and tractors, whether owned or hired, for land preparation. Implements used include ridging ploughs and single furrow mould-boards for the bullocks, whilst disc ploughs and disc harrows are the main tillage implements used with tractors. In most cases farmers just plough and plant, without the harrowing necessary to obtain the good soil tilth essential for good seed-soil content.

Whereas it has not been possible to come out with figures relating to the proportion of lands under bullocks and tractor mechanisation, it can be stated that the proportion of mechanised to total cultivated land remains low.

Planting

Seeding of grains, planting and transplanting of crop are almost exclusively manual operations. Whilst there are some seed drills in the system most small-scale and even some medium-scale farmers establish their crops manually. The large-scale farms however do use mechanical seed drills and planters.

Rice is mostly broadcast, whilst maize, cowpea, groundnut and other crops are sown by manual dibbling, using cutlass, sticks, hoes and at times heels and toes. Some farmers do sow in rows at recommended spacing, using sowing lines and manual dibbling.

Travelling through the country one comes across units of such mechanical equipment as seed drills and precision planters in the farming system, but the adoption of these technologies is not widespread to the extent that even farmers using bullocks and tractors hardly use these seed drills and planters. Except at the Korean Rice Farming School at Asutsuare in the Eastern Region, there was no evidence of transplanting machines. Traditional staples like plantains, yams, cassava, sugar-cane and all the vegetables are planted manually with no mechanical technology interventions.

Weeding and weed control

Weeding is identified as one of the main bottlenecks limiting farmers' ability to expand their farms. It is highly labour intensive and demands critical timing. After planting and crop establishment, and everything being equal, ultimate crop yield depends on the effectiveness of weed control during crop growth.

In Ghana weeding is still predominantly done manually, with the traditional short-handled hoe and the cutlass. Even farmers using bullocks and tractors hardly make use of these facilities for weeding operation. Whilst a few farmers in the mono-cultural systems do use herbicides using knapsacks and boom sprayers, the general level of mechanisation in weed control is very minimal and near zero.

Fertilizer Application

The maintenance of soil fertility on Ghanaian soils has become a necessary requirement for sustained acceptable crop yields. Hitherto, traditional long fallowing of farmlands had help restore soil fertility after periodic cropping. The increase in population with its attendant pressure on land now seem to be shortening the fallow period in most communities, which results in the rapid decline of soil fertility.

In Ghana organic matter is seldom systematically recycled, and very few farmers appear to have ever heard of composts. Except in the Northern and Upper Regions of the country where some level of integration of crop farming and livestock farming exists, the average Ghanaian farmer does not use manures.

The application of inorganic commercial fertilizers has usually been the recommended practice. But whilst a few mechanical fertilizer spreaders and spinners are found with the large-scale commercial farms, the application of fertilizers is generally done manually, either by manual broadcasting, dibbling or application in bands or circles.

The level of mechanisation here is also very low.

Pest and Disease Control

Main use of pesticides in Ghanaian agriculture is by cocoa farmers under a comprehensive Cocoa Marketing Board supported package.

However, a few vegetable and cereal growers, including the small-scale tomato farmers at Akumadan in the Offinso district of Ashanti, do use pesticides and other chemicals extensively. These farmers use manually-operated knapsack sprayers. Large-scale commercial farmers however do use tractor-mounted boom sprayers. All sprayers are imported, with a few of the knapsacks locally assembled.

Harvesting and Threshing

Crop harvesting methods vary depending upon the crop type. Operations are crop specific, and may involve cutting of stalks, stripping, plucking, pulling, digging of roots and tubers, nuts separation of head, pods and ear, collecting, gathering, threshing, shelling, winnowing and other general handling operations.

Hand harvesting with cutlasses, hoes and sickles is common. Attempts continue to be made to introduce improved mechanical technologies into crop harvesting on Ghanaian farms, but the general level of mechanisation is low. Crop-specific technologies may be summarised as follows:

i) Cereals (Rice, Maize, Sorghum and Millet):

Harvesting of cereals on Ghanaian farms is predominantly manual. There have been attempts to introduce such mechanical technologies as reapers, threshers, corn pickers, shellers and combines of varying degrees of sophistication for harvesting rice and maize. Notwithstanding, threshing of rice, sorghum, millet and maize is done manually using hand beating on threshing floors, threshing boxes and stocks. There is limited use of winnowers, cleaners and grading machines.

ii) Roots and Tubers

The Major Roots and tubers crops are yams, cassava, groundnuts, onions, cocoyams and potatoes. Manual mechanical technologies such as groundnut lifters and threshers exist for practically all these crops, but adoption has been slow. In more recent times, there have been attempts to experiment with cassava harvesters, but these are yet to prove their effectiveness.

iii) Plantains, bananas, oil palm, coconut and others

These are tropical crops. The technical problems connected with their mechanical harvesting are yet to solved. They are all manually harvested. The immediate solution to their harvesting lies in improving available hand tools.

iv) Vegetables

All are manually harvested by hand picking or removal with knives and cutlasses.

MECHANISATION CONSTRAINTS IN GHANA

A. Lack of Access in Credit

The most significant barrier to greater agricultural mechanisation in Ghana is the lack of access to credit.

Although some credit is available to farmers for inputs for crop production, the immediate recovery of this at harvest time encourages glut markets and low prices. Quality may also be affected, as in the case of rice being sold and consumed before it has hardened properly. Loans over a longer period, or inventory loans, would allow farmers to get better prices for their crops later in the year, particularly with improved post-harvest treatment and storage. The resulting increase in income would permit investment in labour-saving and value-adding equipment which would in turn generate additional income for the farmers.

The provision of credit to expand primary agricultural production is a major concern for a number of agencies. However, the lack of access to suitable technologies for both crop maintenance and post-harvest processing means that increased yields and increases in cultivated area are likely to exacerbate existing bottlenecks and thus post-harvest losses and poor returns on the investment in inputs. The credit-based investment in crop production should therefore be complemented by corresponding credit support for investment in technologies for activities such as processing, storage and transport. The reduction in post-harvest losses and the increased value which would be added to the crop would then help to finance the mechanisation of the crop production process itself, the expansion of which would then be more viable as the processing bottlenecks were eased.

The dissemination of technologies on the necessary national scale is only likely to occur by means of conventional commercial manufacture and supply mechanisms.

Just as poor credit facilities reduce farmers' ability to improve their economic performance, so the same constraint severely restricts the development of the domestic agricultural engineering industry, which supplies the capital equipment that farmers and processors require.

The potential demand for agro-related equipment is considerably greater than can be supplied by the existing manufacturing organisations that are supported by the government and donors. A significant number of commercial manufacturers of equipment, varying considerably in scale and quality, do exist, but almost without exception the expansion which many of them are attempting is principally financed from revenue, since access to credit is either non-existent or too difficult to pursue. None of the manufacturers interviewed complained about the high rates of interest (possibly because these are academic), only about the lack of credit facilities and the poor performance of those which do exist.

One competent manufacturer's application for a bank loan with which to expand his business was approved, but the mechanisms within the bank proved to be unable to deliver the money. After pursuing credit over a number of years this manufacturer has now given up on the banks, his expansion restricted to what the current profits from his existing business can pay for.

In this way, even when end-users are able to buy equipment, the quantity and quality of what is available is limited, as discussed below. It is even possible to argue that an increase in the number, scale and product quality of manufacturers would eventually lead to an improvement in

consumers' financial access to the equipment, since an improvement in a manufacturer's liquidity through increased earnings may allow the company itself to provide credit to purchasers.

Since a number of agencies are currently providing credit to farmers, albeit principally for crop production inputs and post-harvest processing groups, it is now important to give special consideration to the credit needs of equipment manufacturers also, at every level of scale and sophistication.

However, although it is the major limiting factor on agricultural mechanisation, and thus of the agricultural economy in Ghana, the improvement of access to credit lies outside the Terms Of Reference of this consultancy. It is therefore not appropriate for this report to propose specific recommendations on this subject.

B. Lack of Availability to Agricultural Equipment

The availability of agricultural equipment varies, according to the level of technology.

Traditional hand tools are widely available, hoes and some other tools being made by traditional blacksmiths, while cutlasses, spades and other high carbon or sheet steel products are normally imported. Improved hand-powered land preparation and crop-maintenance equipment is largely unknown, and not normally available. A basic range of draft animal power equipment is available in areas where animal traction has been established. Power tillers (two-wheeled tractors) have been introduced but with only a limited choice of accessories. (Spares for Daedong machines are available only in Tema, but for some other makes there appear to be no spares at all. This appears to occur particularly in cases where the equipment has been introduced as part of a one-off aid package.) The range of accessories used on tractors is also fairly limited and appears to reflect availability as well as high capital cost and the knowledge of the operators. For example, simple chain harrows are not easily available. Accessories and spare parts are more easily available for the better established makes, such as Massey-Ferguson and Ford, but back-up for other brands is sometimes quite inadequate.

Although there is some hand-powered agro-processing equipment in evidence, the choice is limited and the margins of mechanical efficiency over cost are often sufficiently narrow to discourage investment in them by users. Since credit with which to buy equipment with sufficient production capacity for a single family is generally inaccessible for individuals, less attention is given to this market than to the market for motorised equipment which can to be sold to groups (for example via NGO credit) or to commercial operators. Even in cases where such equipment is seen as worth purchasing it is often not introduced in a form suitable for manufacture by informal sector artisans. Thus, since the unit price is too low for the better-equipped small-scale manufacturers, since they lack the necessary distribution network for high-volume production, and the ubiquitous artisans are unable to make it, the product does not become widely available to the people who would buy it. Some of the various forms of hand maize-shellers are examples of this.

It is significant that some intermediate technologies have either been widely adopted or are clearly in the process of becoming so. Two examples of this are the "chorkor" fish smoker, which uses a stack of wire-netting trays over a smoking fire, and the "improved narrow ventilated crib" for maize which is currently being widely disseminated by the Front Line Staff of the DAES, with some apparent success. It would be instructive to examine successful transfers of technology such as these in order to identify the local conditions which allow them to occur

The majority of small-scale powered crop and food processing machinery is now locally manufactured, the designs usually based on existing products or designs from other countries. Quality is very variable, from poorly welded machines made from scrap steel to well-engineered equipment. However, with the exception of just one manufacturer, the consultants saw no equipment which had been marked with the manufacturer's name and location. Thus it is impossible for purchasers and end-users to compare performance and, since cost is invariably a factor, poor quality equipment continues to find a market among the vulnerable. (See 9, below.)

Identifying suitable equipment is further complicated by the difficulty in researching what is available and who makes it. There is one directory of equipment, published by the DAPIT project (Development and Application of Intermediate Technology), but this is limited in geographical scope and the number of manufacturers represented. Since one of DAPIT's objectives is to encourage Ghanaian industry, the directory excludes imported equipment, thus severely reducing the information available to purchasers and users. In a number of cases, organisations expressing a need for equipment of a particular type were unaware that it already exists elsewhere in Ghana.

The better small-scale manufacturers of agricultural and processing equipment are all attempting to expand in spite of the lack of credit (see 7.1, above), and assert that there is a market for considerable increases in their production. However, almost no marketing is carried out by the private sector. The manufacturer referred to earlier, whose agreed loan was not delivered, advertised on one occasion but was so overwhelmed by the response that he has not dared to do so again, since his current manufacturing capacity is insufficient to meet the potential demand.

The combination of the poor support of imports and the lack of information regarding sources and quality means that end-users have little choice or knowledge and are therefore at risk, the frequent examples of which are a severe discouragement to others to invest in mechanisation. Since the investor has no way to tell the difference between good and bad machines she or he is likely to adopt a very cautious attitude to all such equipment, which considerably retards mechanisation. This applies both to imported and domestically-manufactured equipment. The unavailability of guidance on the suitability of different equipment according to its proposed deployment further increases the risk to the investor.

Equipment is restricted in choice, difficult to locate, of very variable quality, and often with poor back-up. Considering the combination of the supply situation with the difficulties of access to credit, it is not surprising that private investment in mechanisation technologies continues to be poor. Much of what does occur depends upon the provision of credit by NGO's, which is inevitably selective and restricted by resources.

C. Lack of Access to Information

Information for consumers directly affects their access to equipment and its effective availability, as discussed above.

Access to information is also crucial for manufacturers. Market research and product development are expensive and are consequently generally beyond the means of small businesses. Although the term "design" is used very freely, most farming and agro-processing products are copies or variations of existing designs. In most cases the source of these designs is a machine made by someone else, usually fairly locally. Thus the range of options available to consumers remains very limited, and the performance of the machine is likely to be degraded during the copying process. Enquiries concerning the design sources of the better manufacturers revealed

that most use only one or two sources. The private sector most commonly use one of the Intermediate Technology Transfer Units (ITTUs), while AESD engineers may also have a particular book or similar source to which they refer. However, the ITTUs themselves use very limited sources of information. GRATIS has an official link with ISAT, the information service of GTZ, but no reference was made to using this by any of the ITTUs interviewed. The foreign source most mentioned by official agencies and NGOs was the International Institute of Tropical Agriculture (IITA) in Ibadan, Nigeria.

A number of international aid agencies and NGOs have produced a considerable number of books containing detailed practical information on a wide variety of technologies, but limited financial resources prevent access to this rich source. Surprisingly, examples of such books were not even in the hands of DAPIT or of those ITTUs whose libraries were inspected. A number of useful books containing equipment designs are available from the libraries of the British Council in Accra and Kumasi, but none of the manufacturers or agencies interviewed referred to these as a source of information.

Visits to a variety of manufacturers revealed a number of recognised product opportunities and a requirement for information on designs with which to respond to them. There is also a considerable potential demand for designs of metalworking and other equipment which could be built for use within the workshops themselves.

The flow of technical and product design information within Ghana is very poor. The answers to problems which were presented in one place were frequently answered in another within the country. Some attempts have been made by NGOs to share information but these have largely been unsystematic.

The flow of information on technologies and equipment developed and used in other countries is also very poor, in spite of the existence of aid agencies dedicated to the provision of such information, for example ITDG, ISAT, TOOL, GRET, FIT (ILO & TOOL), ATOL, SKAT etc.

It should be stressed that a number of agencies, government, NGO and commercial, are working hard to introduce and disseminate technologies appropriate to Ghana's needs. However, there is no discernable systematic monitoring of the performance of the equipment concerned, no comparative evaluation and little knowledge of the experiences of others. Isolated experience leads to the repetition of the same experiments and mistakes, with very little of the communal learning necessary for more rapid development. The lack of systematic monitoring and the sharing of the information which it would generate affects both development agencies and end-users.

The lack of information on the performance of technologies which have already been deployed in the country applies generally, both to imported and locally manufactured equipment.

The lack of provenance marking on locally-manufactured equipment means that, even if there were a monitoring system, it would be impossible to identify who manufactured a particular example of a product. If the manufacturer of a poor or dangerous machine cannot be identified then it is impossible to take remedial action to assist that manufacturer to improve the quality of his or her products. At the moment there is no system for the identification of inadequate or dangerous equipment.

The quality of imported equipment also varies considerably, usually in direct proportion to cost.

A variety of stationary diesel engines have been imported commercially and are widely deployed to power small processing equipment of various kinds. A significant proportion of this equipment is acquired through credit provided by aid agencies, often to women's groups, which must be repaid. Since capital cost is a major factor, even with credit, purchasers with no prior mechanical knowledge are likely to buy less costly machines. (The difference in cost between makes may be as high as a factor of three.) Since purchasers and the people whom they delegate to operate their machines are frequently inexperienced and poorly trained, if at all, in servicing and maintenance, it is easily assumed that the repeated and sometimes rapid failure of the diesel engines is due to operator failure and a lack of maintenance. However, the consultants noticed that, in the mills which they visited in which the operators complained of mechanical breakdowns, a particular make of stationary diesel engine occurred several times. Was this a coincidence, or is it a poor product which prospective purchasers and credit-providers should be warned about? This can only be established by the systematic monitoring of equipment which is already in commercial use in the country.

Every failure of a capital investment, particularly by resource-poor groups, whether it is due to lack of training or poor equipment, further discourages such investments and thus retards the widespread adoption of such technologies. The recording of experiences and the availability of that information can reduce this problem.

Farming, Origins of Farming

I. INTRODUCTION

Farming, Origins of, the beginning of the cultivation of plants and the domestication of animals in prehistory. The origins of farming are generally equated with the Neolithic period, or Late Stone Age, and generally denote a move away from the nomadic lifestyle of the hunter-gatherer. They are associated with the adoption of a sedentary way of life, the development of permanent settlements, and the first occurrence of pottery vessels for cooking and storage. The origins of farming, also sometimes referred to as the Neolithic revolution, occurred about 10,000 years ago in the Near East, 8,000 years ago in China, and probably a few thousand years later in the New World.

II CULTIVATION AND DOMESTICATION

The development of farming has traditionally been seen as a revolution, a new way of life made possible by the sudden adoption of cultivation and stock-rearing by enlightened hunter-gatherers. In recent decades, however, it has become apparent that hunting, fishing, and gathering generally represent a far easier and more leisurely way of life, while farming requires considerable toil in tilling the soil, sowing, weeding, and harvesting. Farming, therefore, was not a superior way of life but was probably adopted only out of necessity. It is now generally assumed that quite specific factors influenced this development: an increase in population made possible by improved environmental conditions at the end of the Ice Age, or climatic hardship, or dwindling opportunities for hunting and gathering have been cited as contributing to this development.

The origins of farming constitute a gradual evolution rather than a sudden revolution, though, when seen against the immense timescale of prehistory, the new way of life was adopted relatively quickly. However, it is significant that the plants and animals chosen for domestication were not staple resources at the inception of farming; instead, people seem to have become economically dependent on them over many centuries, and in some cases millennia, after the initial stages of domestication. Thus the many stages of ““semi-domestication””, the cumulative changes in the relationships between humans and natural resources, must be recognized. Farming replaced foraging only very gradually, and in many parts of the world hunting, fishing, and gathering persisted long after the introduction or adoption of domesticated resources. It is, therefore, impossible to pinpoint any time or place at which the domestication of any species began. In this sense the ““origins of farming”” cannot accurately be identified.

Hunter-gatherers were certainly not ignorant of the principles behind agriculture and stock-rearing. They would have known, for example, that seeds planted at certain times of year would produce crops that could be harvested, and indeed they may have done this occasionally themselves. They would often have taken action to safeguard animal resources, avoiding killing very young animals or pregnant females, for example. But being essentially mobile, hunter-gatherers were not tied to a sedentary life or to the year-round tending of crops.

Crucial to the issue of identifying how, when, and where fully fledged farming came into existence is the status of the species in question: that is, whether the remains of plants and animals that suggest domestication in the archaeological record were indeed domesticated. With regard to plants, the distinction between wild and cultivated forms is not always clear and, even when a difference is apparent, the length of time involved in this divergence is unknown. Cultivated

cereals are quite easily distinguished from their wild counterparts, however; clues includes the loss of the brittle rachis (stem) that facilitated the dispersal of seed by natural agents: this change in morphology is the direct result of the deliberate development of varieties that retained their seeds until they could be harvested. Recent measurement of domestication rates in wild wheats and barleys under primitive cultivation suggests that the transition from wild to domestic could in fact have been complete within only 20 to 200 years.

Where animals are concerned, one criterion of domestication is selective breeding, leading to a distinction between the physical characteristics of wild and domesticated species. This distinction is usually measured in terms of changes in bones and teeth, such as a reduction in jaw size. But specialists disagree about which physical changes indicate domestication, while emphasis on the wild/domestic polarity masks a whole spectrum of human-animal relationships such as taming, loose control, herd following, or herd management without selective breeding. Moreover, as with plants, we have no idea how long such changes took to become apparent after the establishment of domestication.

Wild herbivores are remarkably easy to control, and hunter-gatherers, being intimately acquainted with the habits of all animals, would certainly have been capable of controlling or even corralling them. There are a few clues to some kind of control of animals in the Ice Age, including a few depictions of what seem to be simple horse bridles in France, about 14,000 years ago; and it is certain that the dog was domesticated in Eurasia in the same period, presumably for companionship and protection, and as a hunting aid.

III ANIMALS

In the Middle East, sheep and goats seem to have been domesticated by 9,000 years ago. As a source of meat they gradually replaced the gazelle, which had long been hunted in the region and may itself have been herded. Sheep and goats were probably chosen for domestication since they could graze on a wider range of plants, so that the land could support more of them. These domestic animals were gradually introduced into Europe and spread along the Mediterranean shores during the Neolithic period. Pigs may have been domesticated in Turkey and China several centuries earlier; in their domesticated form, smaller and more docile than the wild boar, they then spread through Eurasia.

Recent studies of DNA (*see* Nucleic Acids) taken from present-day cattle on three continents have overturned the idea that their domestication spread from a single centre in the Near East; instead, biologists have found evidence that the domestication of wild oxen occurred in at least two separate locations: in south-west Turkey and east of the Iranian desert, with a possible third site in Africa. Since the genetic differences between the groups were found to be too great for them to share a common ancestor 10,000 years ago, they must result from independent domestications of different races of wild ox. Cattle pastoralists did not become established in southern Africa until AD 1000, perhaps because the local peoples had a biological intolerance of lactose (sugar present in milk).

Sheep and goats, like cattle, were first domesticated for meat, with milk and wool becoming important somewhat later as the animals were bred for these features. In fact it has been suggested that in some areas of the Old World, after about 3500 BC, a marked change occurred: domestic animals were no longer exploited solely for the primary products of meat and hides but also for secondary products such as milk, cheese, and wool, and they were used as beasts of burden: archaeological evidence for this is supported by carvings and reliefs on seals from Mesopotamia that show ploughing, milking, and carts (that are assumed to have been drawn by oxen). This change, like the initial origins of farming, has been attributed to further population growth and territorial expansion, when people found it necessary to penetrate more marginal environments and to exploit livestock more intensively. Evidence from central Europe, concerning the age and sex of cattle deduced from bones found in settlements, and what appear to be pottery cheese sieves, indicates that dairy farming was taking place here as early as 5400 BC.

In the New World, there were fewer potential domestic animals, but llamas and alpacas were domesticated by 5400 bc for their coats and as pack animals, as well as for food; guinea pigs were raised for meat in South America, turkeys in Mexico; and dogs, which were domesticated in the New World as early as in the Old World, seem to have been used primarily as a source of meat.

IV PLANTS

Wheat, barley, and legumes such as peas and lentils were first domesticated in the Near East and then spread through Europe. At the end of the Ice Age some large-seeded grasses, the ancestors of modern cereals, began to grow in the hills in this region. The waterlogged site of Ohalo II, by the Sea of Galilee in Israel, has yielded the world"s oldest-known cereal grains: the charred remains of hundreds of grains of wild wheat and barley dating to 19,000 years ago, together with many other plants and fruits, and a rich faunal assemblage indicating a broad-spectrum economy of fishing, hunting, and gathering. Israeli archaeologists argue that the harvesting of cereals has deep roots in Natufian times (12,500-10,000 years ago), and that this harvesting gradually intensified into deliberate cultivation. Deposits at Jericho and elsewhere contain evidence of the cultivation of cereals and legumes by the end of that period, and this is confirmed by microwear on stone tools, suggesting the small-scale cultivation of wild-type cereals in the Jordan Valley. Hunter-gatherers became increasingly dependent on these nutritious, easily stored grains, harvesting them with flint knives and grinding them to flour. Presumably, they began to scatter some grain to improve the extent or density of the plants" growth, and later sowed the cereals outside their natural habitat. By 10,000 years ago, domesticated forms had emerged, with larger grains, and a tougher seed-head that did not shatter on ripening and so prevented the ears being lost before threshing. The olive, which became one of the most important plants in the economy of the Mediterranean basin, was probably already being cultivated by about 8,000 years ago.

Grinding stones and sickle blades from 18,000 to 11,000 years ago in the Nile Valley indicate intensive gathering and use of plant foods. Indigenous species such as sorghum and millet (and possibly African rice) were under cultivation in North Africa by 6000 bc (though the times and places of their initial domestication are still uncertain) while wheat and barley were introduced from western Asia around this time. They spread down the Nile Valley, together with the use of the plough, but could not penetrate the tropics where tubers and tree crops were brought into cultivation.

Rice, millet, cereals, and root crops were cultivated in South and East Asia. Millet and cabbage were being grown in northern China by people living in villages dating to 8,500-7,000 years ago.

Barley may have been domesticated in China independently from western Asia, before 4,000 years ago, and spread to Korea and Japan during the next millennium. The specific places where the cultivation of rice began are still undetermined. It is very difficult to differentiate between wild and cultivated rice, but wet-rice farming appears to have been well established in southern China by 7,000 years ago, implying that the initial stages in its domestication probably occurred much earlier. It spread into northern China by 5,000 years ago, but did not reach Korea and Japan until about 1,000 years later.

Pollen and other botanical evidence suggests that maize appeared in South America over 8,000 years ago. There is, however, no solid archaeological evidence for its cultivation in the New World until 5,600 years ago. One early centre of cultivation was the Tehuacáán valley, in Mexico, where more than 100 species of plants, including maize, beans, squashes, potatoes, tomatoes, and manioc (Cassava) were grown. Maize cobs dating from this early period are small, but grow steadily in size in later periods. It is thought that its wild ancestor was teosinte, a grass native to parts of Mexico, but the original hearth of maize domestication has yet to be found. Squashes, avocados, potatoes, peppers, beans, and gourds may also have been under cultivation before maize - some of them perhaps even 10,000 years ago - but took some time to become important in the diet.

HUNTER-GATHERERS

I. INTRODUCTION

Hunter-Gatherers, people practising subsistence techniques of hunting and gathering wild food, with little or no agriculture. In many parts of the world small groups of hunter-gatherers still exist. These are peoples with a known and acknowledged recent history of living largely by hunting wild animals and birds, by fishing, by gathering wild fruits, nuts, and mushrooms, by digging up edible roots and tubers, and by extracting honey from the nests of wild bees. In the rapidly changing world of today these activities rarely provide even as much as 50 per cent of the diet of the people concerned, but typically remain highly valued and a central focus for group identity.

II HUNTER-GATHERER GROUPS

Among the best-known of these groups are the Aboriginal peoples of Australia and the Inuit (formerly known as Eskimo) peoples of Greenland, Canada, Alaska, and northern Siberia. Most of these groups today obtain the greater part of their subsistence from welfare payments and wage labour. The San (Bushmen) of Botswana, Namibia, and southern Angola have lost most of their land and now mostly live as low-status landless labourers among the people who have taken it. Many thousand Pygmies in the tropical rainforests of Zaïire, Congo, the Central African Republic, Cameroon, and Gabon are still active hunters but gather little vegetable produce, preferring, as they have throughout their known history, to obtain vegetable foods by trading meat with and working for their agricultural neighbours.

There are many less well-known hunter-gatherer or former hunter-gatherer groups in Somalia, Ethiopia, Kenya, Tanzania, Rwanda, and Burundi; in Canada, the United States, and a number of South American countries; in Russia; and in India, Thailand, Malaysia, Indonesia, and the Philippines. All are today finding it difficult to continue to live what is in almost every case their preferred way of life by hunting and gathering, and are under great pressures from governments and from their various neighbours to surrender their lands, to hand over their children to school systems which alienate them from their own cultures, to accept a fully sedentary way of life, and to repudiate aspects of their own culture which those who have political power over them find distasteful. Indigenous peoples movements are struggling against these overwhelming predatory pressures, but have in most places so far made little headway. The ever-present danger is that groups dispossessed of their chosen way of life, of their land, of their sense of their own value and identity, may end up as impoverished and stigmatized welfare recipients or as beggars and prostitutes.

II RESEARCH

In recent years much research has been carried out among these various hunter-gatherer and former hunter-gatherer groups. This research is important for three main reasons.

First, it is contributing to growing public awareness of the appalling history of human rights violations in the treatment of these groups. It is providing data which, in a small number of cases, has already helped such groups to secure legal title to at least part of their lands.

Secondly, such research has sought to provide possible clues to the varied ways of life of human

societies during the later phases of the million or two million years of human existence before the first development of agriculture and pastoralism some 10,000 to 15,000 years ago. For more than 99 per cent of human history, people lived exclusively by hunting and gathering. By the end of this vast period, people had developed all of the crucial capacities, physical and intellectual, which are shared by all the world's present-day human populations.

The relevance of modern hunter-gatherers for understanding the hunter-gatherers of the pre-agricultural period is a highly controversial issue. Human societies are consistently innovative, constantly generating changes in their living arrangements, cultures, and systems of values. Internally generated changes have been supplemented by changes induced by political and economic pressures, and opportunities deriving from ever-increasing contacts with politically and economically dominant outsiders.

The question is how far the effects of these changes have been limited by the commitment to the hunting and gathering way of life. The very striking resemblances in the types of social organization found among hunter-gatherers in different parts of the world with very different histories strongly suggest that the fact of living by hunting and gathering has certain implications for social organization which are not eroded by internally or externally generated social change. If this is correct, the organizational forms found among modern hunters may well have also existed in the pre-agricultural period, possibly supplemented by additional now-extinct forms of organization. Some researchers would accept and others would reject this suggestion.

The third reason why research among modern hunter-gatherers is important is a particularly interesting one. Everywhere in the world today politicians and ordinary people, whatever their political views, are concerned about the issues of equality and inequality. Some - not all - present day hunter-gatherers are the least socially differentiated societies in the world today and provide important data for those seeking to understand human social forms. Differences of power, wealth, and prestige between individuals are less than in any other societies. The evidence suggests that such equality is not a result of past or present food shortages or more generalized poverty. Instead it is linked with a political ideology which stresses the moral value of sharing property widely among all in the local community, and not accumulating it for the exclusive use of the owner and his or her family and other close associates. Among these societies is found the greatest known equality between men and women, between old and young, between parents and children, between leaders and followers, between religious experts and religious congregations, between the skilled and the unskilled, between the strong and the weak. Whatever their history, these cases provide a kind of baseline of fundamental human social forms. A detailed understanding of the operation of these social systems is directly relevant to the task of understanding and explaining the operation of the vast range of more differentiated societies in which the rest of the world's peoples live.

Hunting in the Kalahari Desert

Formerly called the Bushmen by white settlers, the ethnic groups known as the San and Khoikhoi inhabit the Kalahari Desert in southern Africa. They have lived in the hot and dry territory of this mostly red-soiled desert for thousands of years. They subsist as nomads among the sparse vegetation of the region by gathering wild plants and fruit and by hunting animals for meat.

AE 162 - BASIC AGRICULTURE I (2,0,2)

Dr. Joseph Ofei DARKO, BSc (Hons) Agriculture, MSc, PhD (Agric. Engineering)

Crop Production

1. Principles and practices of arable and plantation crops.
2. Factors affecting growth and production of crops.
3. Soil/plant/water relationships.
4. Technical and economic aspects of farming systems.
5. Agronomic/farming practices and principles covering them.

Animal Production

1. The field of animal production, feeding, breeding.
2. Introductory poultry science.
3. Animal production mechanization.